



CHAPTER TWO

Basic Principles & Standards

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Introduction



- Drawing standards are **set of rules** that govern how technical drawings are represented.
- Drawing standards are used so that drawings **convey the same meaning** to everyone who reads them.

Standard Code

Country	Code	Full name
Thailand	มอก./TIS	สำนักงานมาตรฐานผลิตภัณฑ์อุตสาหกรรม
USA	ANSI	American National Standard Institute
Japan	JIS	Japanese Industrial Standard
UK	BS	British Standard
Australia	AS	Australian Standard
Germany	DIN	Deutsches Institut für Normung
	ISO	International Standards Organization

Examples of JIS Standard

Code number	Contents
JIS Z 8311	<i>Sizes and Format of Drawings</i>
JIS Z 8314	<i>Scales</i>
JIS Z 8312	<i>Line Conventions</i>
JIS Z 8313	<i>Lettering</i>
JIS Z 8315	Projection methods
JIS Z 8316	Presentation of Views and Sections
JIS Z 8317	Dimensioning

Drawing Sheet : Standard size

■ Trimmed paper of a size A0 ~ A4.

■ Standard sheet size
(JIS)

A4 210 x 297

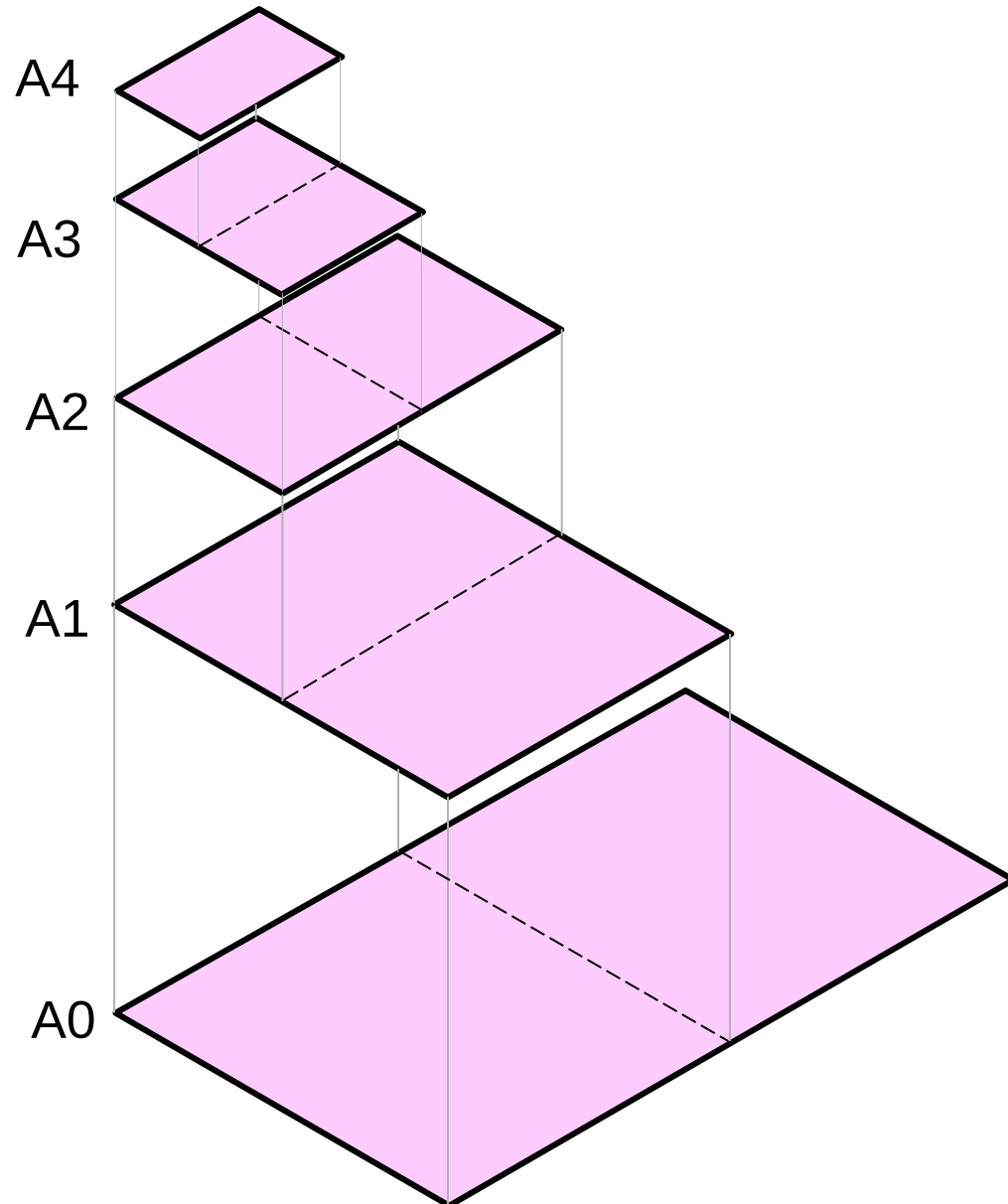
A3 297 x 420

A2 420 x 594

A1 594 x 841

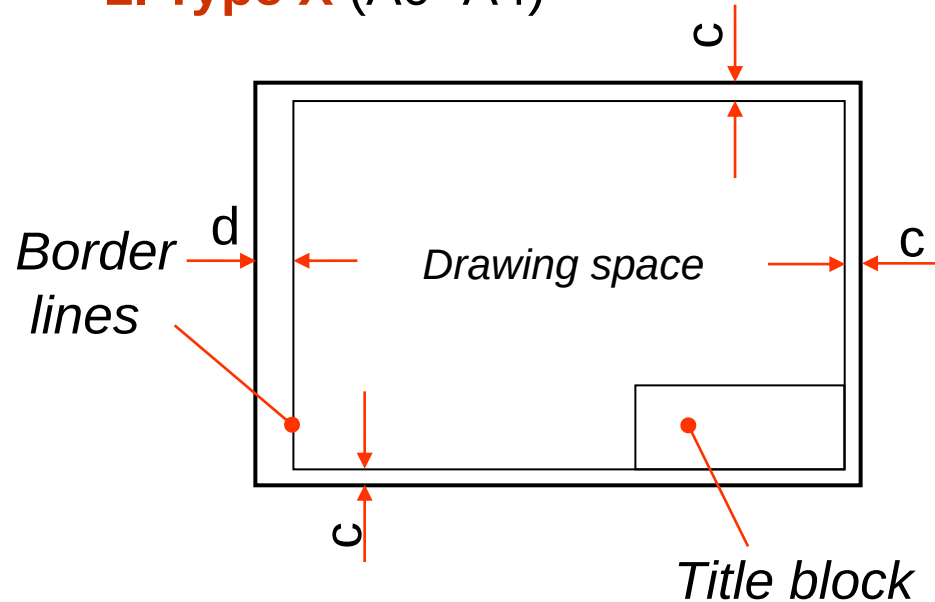
A0 841 x 1189

(Dimensions in millimeters)



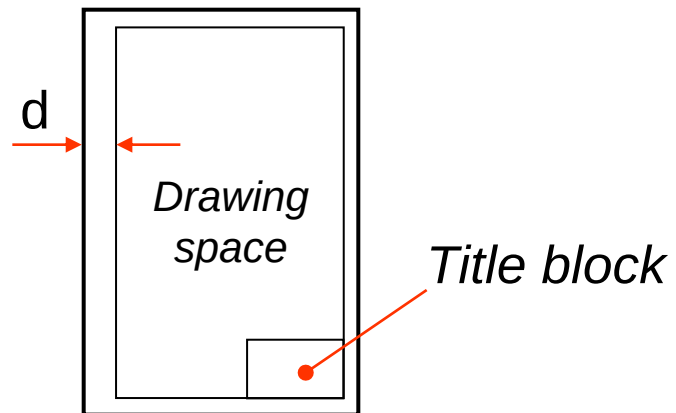
Drawing Sheet : Orientation & Margin

1. Type X (A0~A4)



Sheet size	c (min)	d (min)
A4	5	20
A3	5	20
A2	5	20
A1	5	20
A0	5	20

2. Type Y (A4 only)

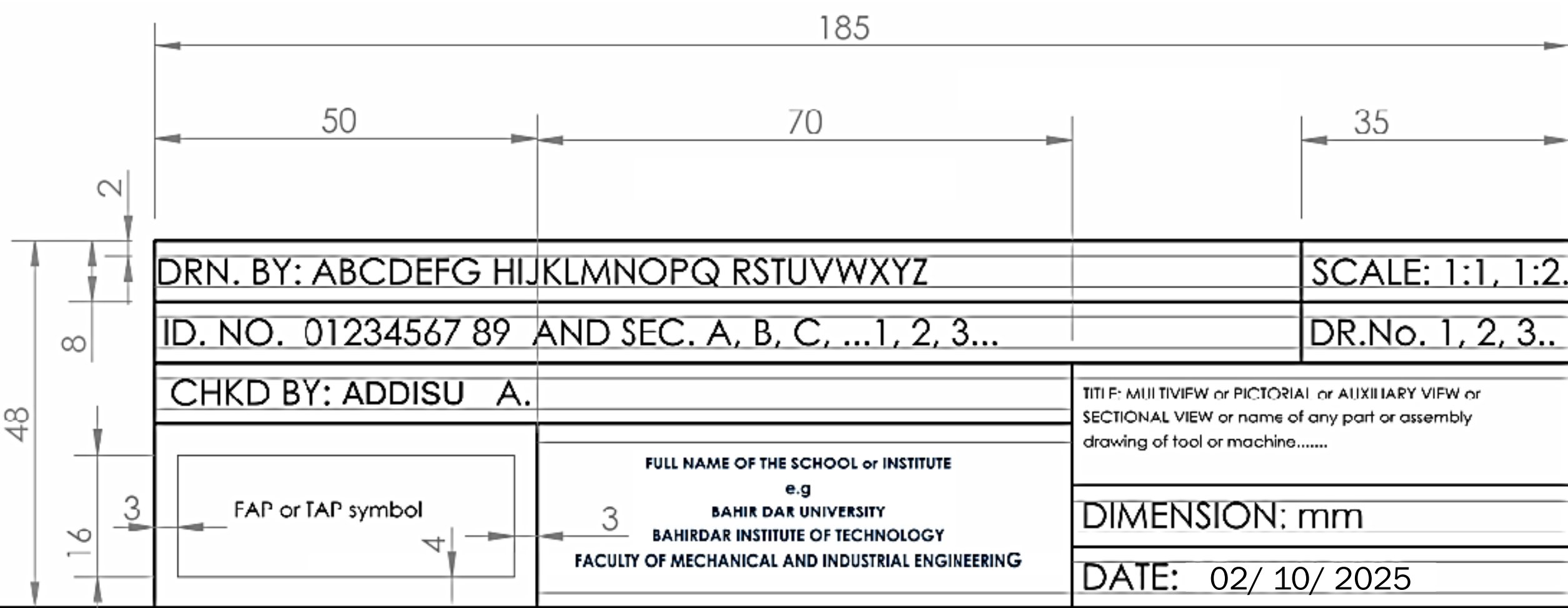


Contents:

- I. Title of the drawing
- II. Sheet number
- III. Scale
- IV. Symbol, denoting the method of projection
- V. Name of the firm
- VI. Initials of staff drawn, checked and approved.

Drawing Sheet : Orientation & Margin

Cont'd



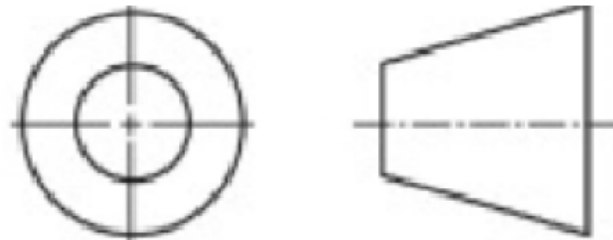
Drawing Sheet : Orientation & Margin

Cont'd

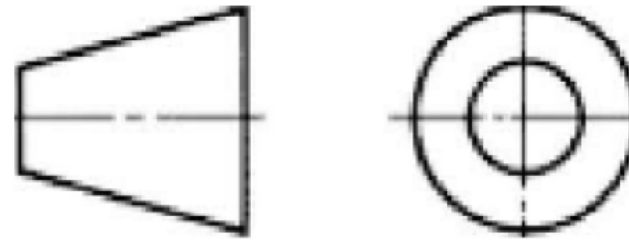
This title block is constructed on the landscape orientation of A4 size paper. If the given drawing is too large, you can construct using an appropriate scale; but you must check both the portrait and landscape paper orientation whether the net drawing area is sufficient or not.

NB: The dimensions of the title block is the same for all paper size and paper orientation unless you are told to change so.

Third angle projection (TAP)

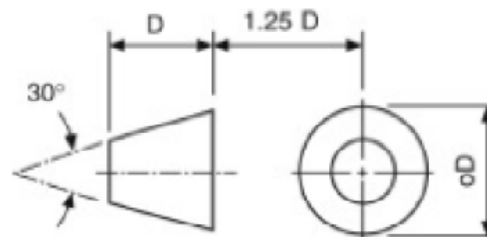


First angle projection (FAP)



Dimensions how to construct FAP or TAP representations

D=16mm



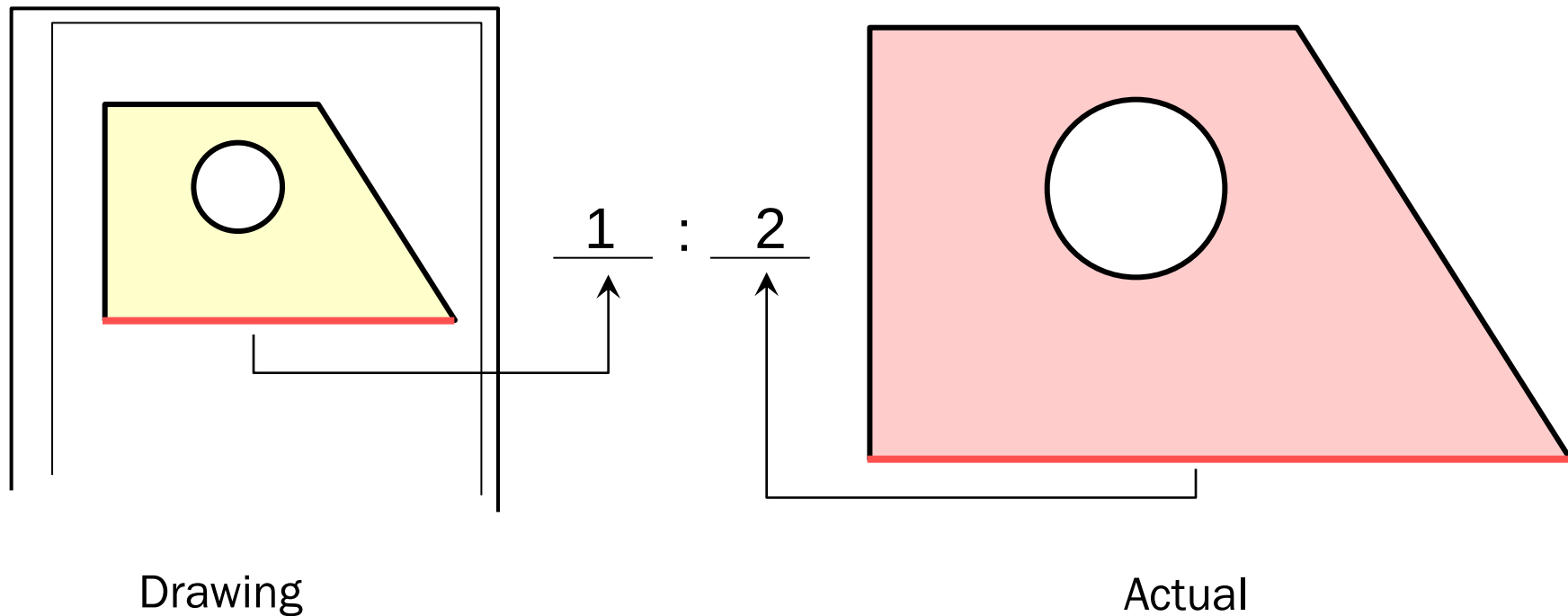
paper margin

border line

Drawing Scales : Definition

Length, size

- **Scale** is a ratio between the linear dimension of a drawn representation of an object and the actual object.



Drawing Scales : Designation

- Designation of a scale consists of the word “SCALE” followed by the **indication of its ratio**, as follows

SCALE 1:1 for full size

SCALE **X**:1 ($X > 1$) for an **enlargement** scales

SCALE 1:**X** ($X > 1$) for a **reduction** scales

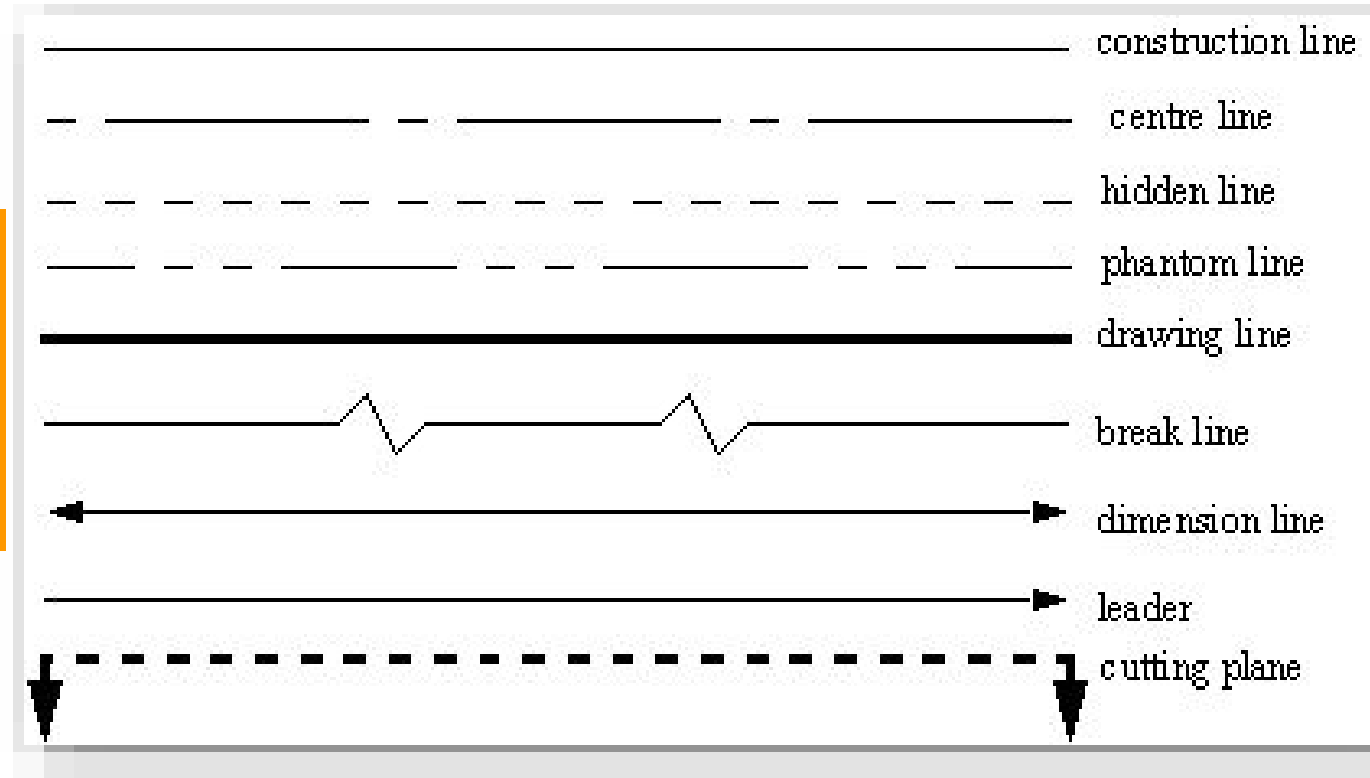
- Drawing scale is commonly found in a title block.

Drawing Scales : Notes

- **Dimension numbers** shown in the drawing represent the “true size” of an object and they are independent of the drawing scale used.
 - Standard reducing scales are
1:2, 1:5, 1:10, 1:20, 1:50, 1:100
 - Standard enlarging scales are
2:1, 5:1, 10:1, 20:1, 50:1, 100:1
- Standard scale

Drawing Standard

Line Types



Basic Line Types & Name according to application

Thickness Style	Thick	Thin
	Visible line	1. Dimension line 2. Extension line 3. Leader line
Continuous		
Dash		
Chain		

Hidden line

Center line

- 1. Visible line** represent features that can be seen in the current view.
- 2. Dimension line**
Extension line indicate the sizes and location of features.
Leader line
- 3. Hidden line** represent features that can not be seen in the current view.
- 4. Center line** represents symmetry, path of motion, centers of circles, axis of axisymmetrical parts

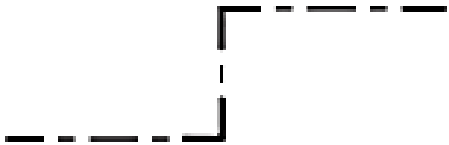
Basic Line Types

Cont'd

<i>Line</i>	<i>Description</i>	<i>General Applications</i>
A 	Continuous thick	A1 Visible outlines
B 	Continuous thin (straight or curved)	B1 Imaginary lines of intersection B2 Dimension lines B3 Projection lines B4 Leader lines B5 Hatching lines B6 Outlines of revolved sections in place B7 Short centre lines
C 	Continuous thin, free-hand	C1 Limits of partial or interrupted views and sections, if the limit is not a chain thin
D 	Continuous thin (straight) with zigzags	D1 Line (see Fig. 2.5)

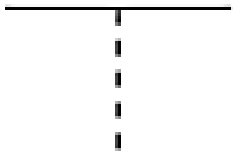
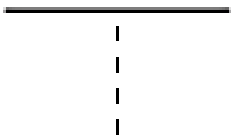
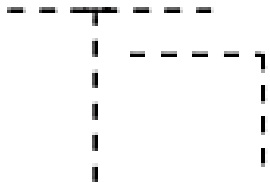
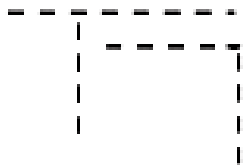
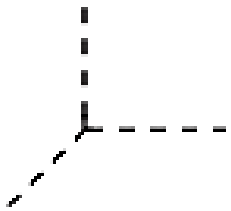
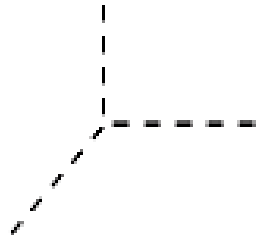
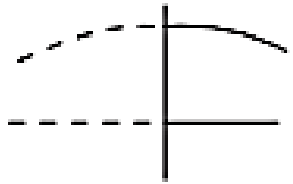
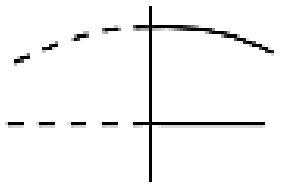
Basic Line Types

Cont'd

E 	Dashed thick	E1 Hidden outlines
G 	Chain thin	G1 Centre lines G2 Lines of symmetry G3 Trajectories
H 	Chain thin, thick at ends and changes of direction	H1 Cutting planes
J 	Chain thick	J1 Indication of lines or surfaces to which a special requirement applies
K 	Chain thin, double-dashed	K1 Outlines of adjacent parts K2 Alternative and extreme positions of movable parts K3 Centroidal lines

Common Mistakes

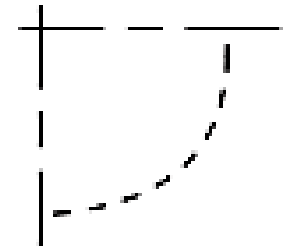
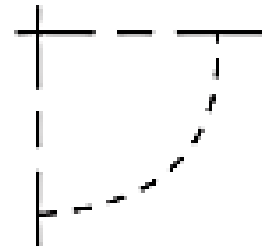
Cont'd

<i>Instructions</i>	<i>Correct</i>	<i>Incorrect</i>
Begin with a dash, not with a space		
Dashes intersect without a gap between them		
Three dashes meet at the intersection point		
As a continuation of a visible line/arc, begin with space		

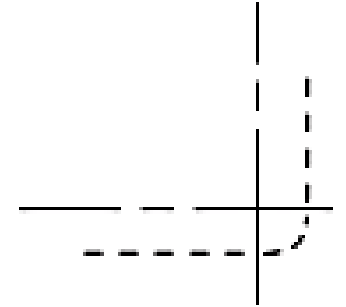
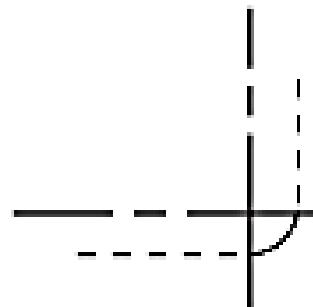
Common Mistakes

Cont'd

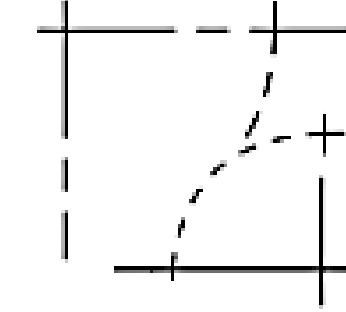
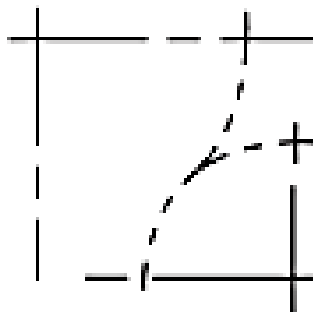
Invisible arcs begin with a dash



Small arcs may be made solid

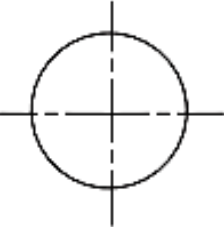
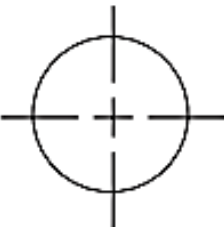
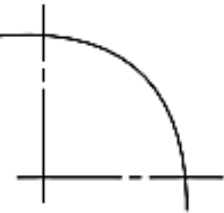
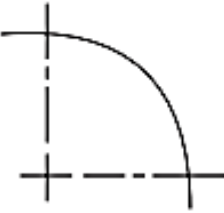
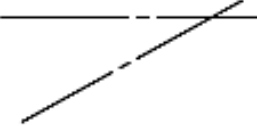
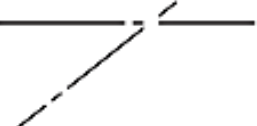
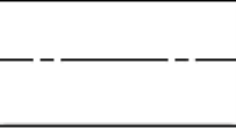


Two arcs meet at the point of tangency



Common Mistakes

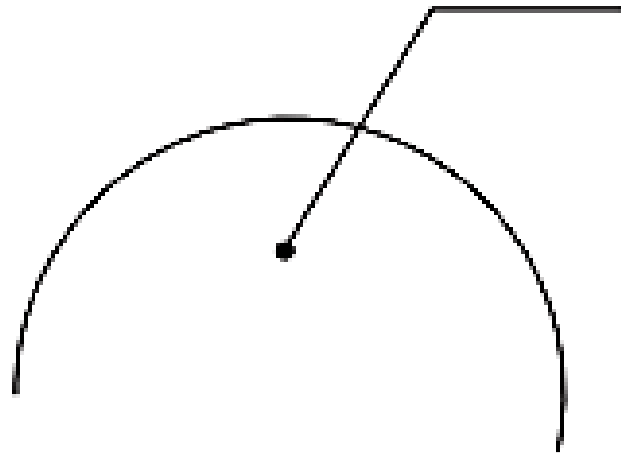
Cont'd

<i>Instructions</i>	<i>Correct</i>	<i>Incorrect</i>
Axis line starts and ends with a longer dash		
Two axes intersect with longer dashes		
		
		
		

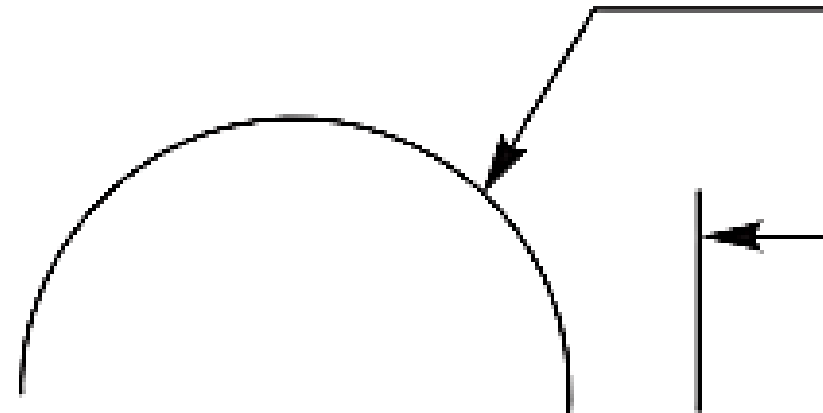
Leader Line

Cont'd

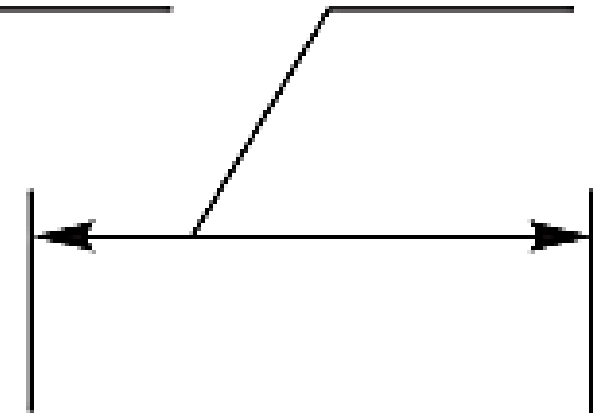
1. With a dot, if they end within the outlines of an object,
2. With an arrow head, if they end on the outline of an object,
3. Without dot or arrow head, if they end on a dimension line.



(a)

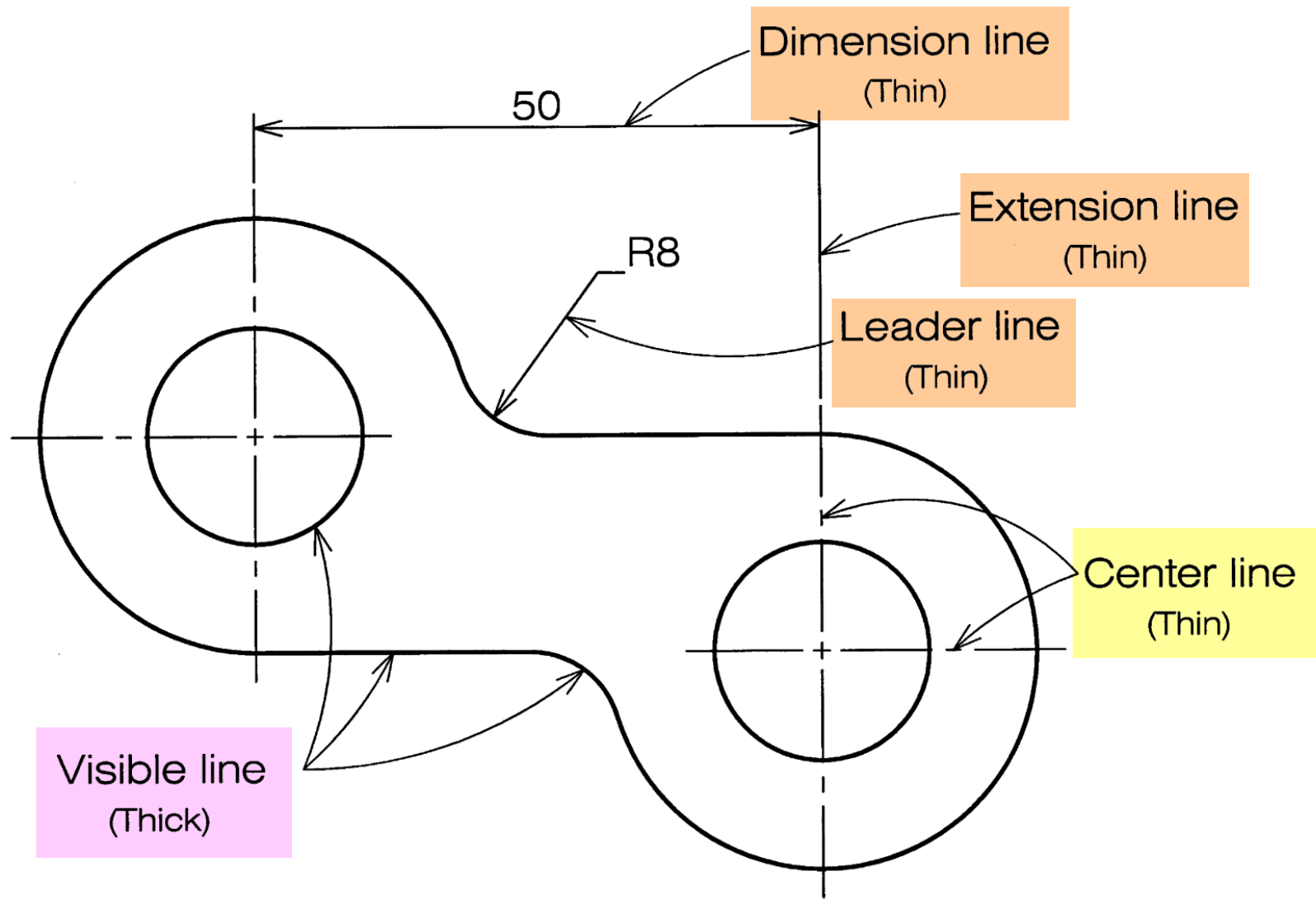


(b)

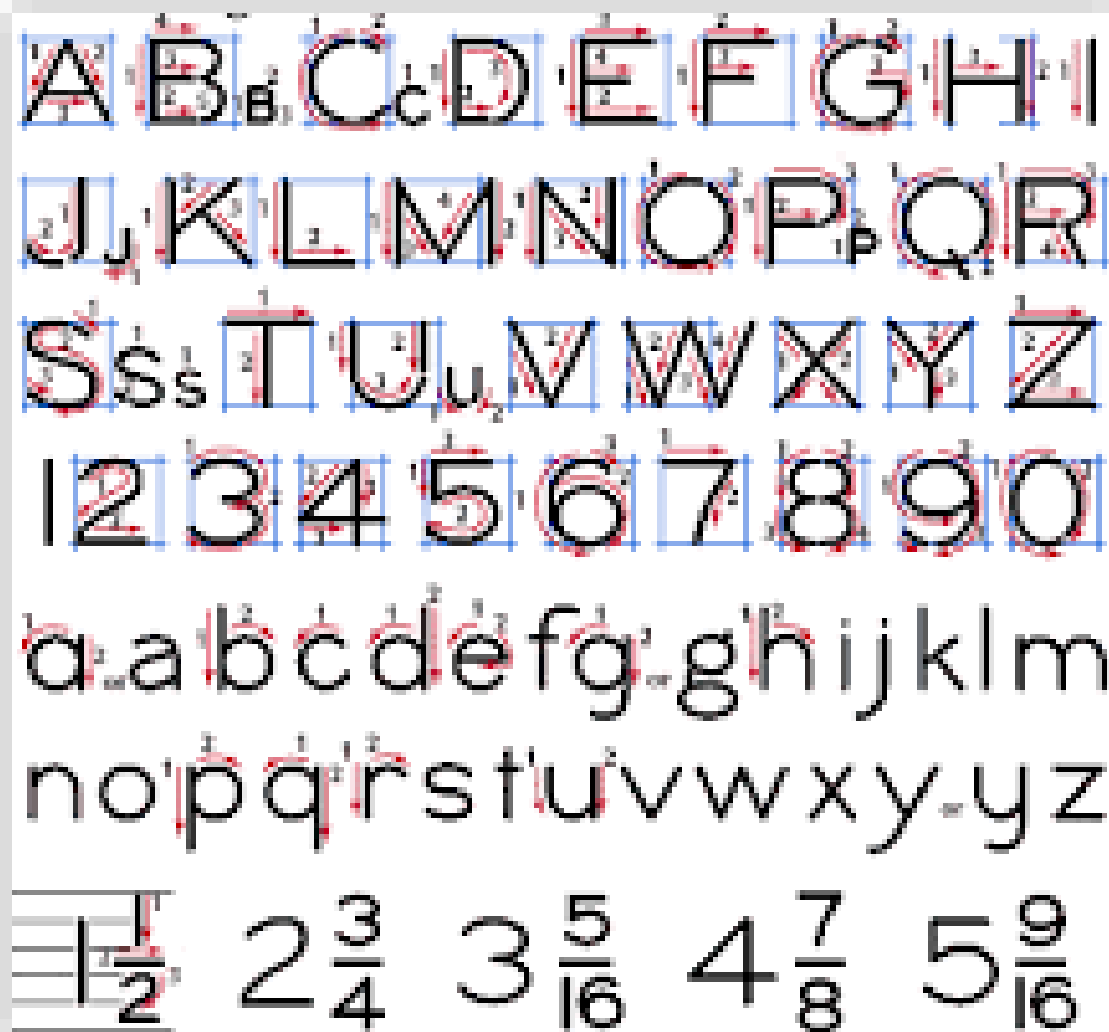


(c)

Example



Lettering



Recommendation

- Text's style on the drawing must have the following 2 properties

Legibility

- Shape
- Space between letters
- Space between words

Uniformity

- Size (or text height)
- line thickness

Examples

ESTIMATE **GOOD**

ESTiMaTE Not uniform in style.

ESTIMATE
ESTIMATE Not uniform in height.

EST/MATE
ESTIMATE Not uniformly vertical.

ESTIMATE
ESTIMATE Not uniform in thickness of stroke.

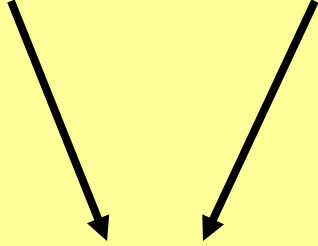
ESTIMATE Inappropriate space between letters

Basic Strokes

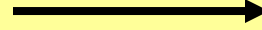
Straight



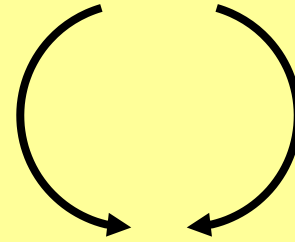
Slanted



Horizontal



Curved

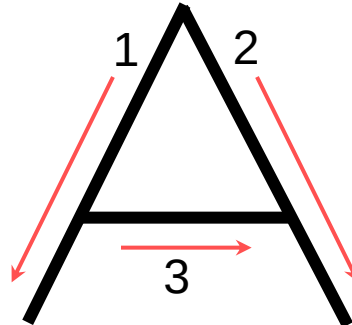


Examples

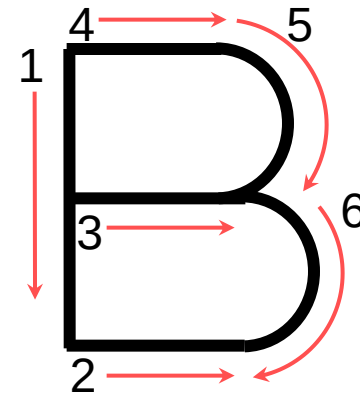
“I” letter



“A” letter



“B” letter

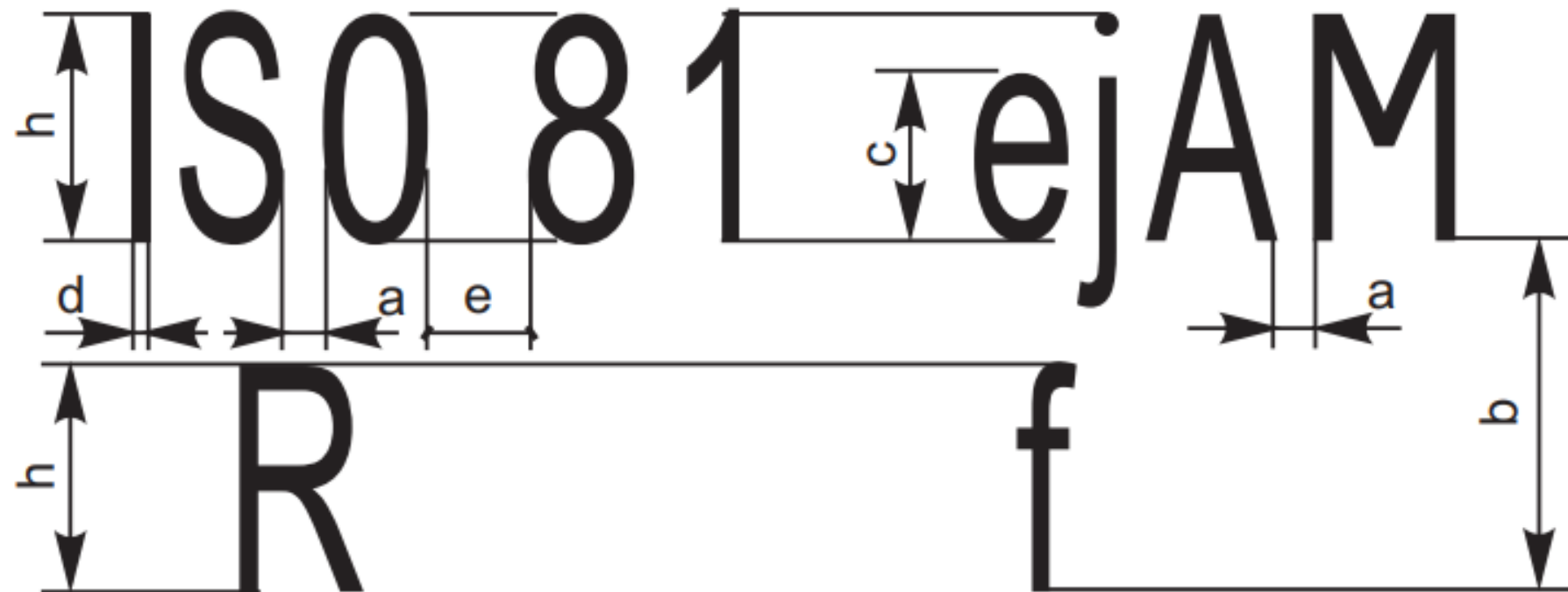


Lettering Style

Cont'd

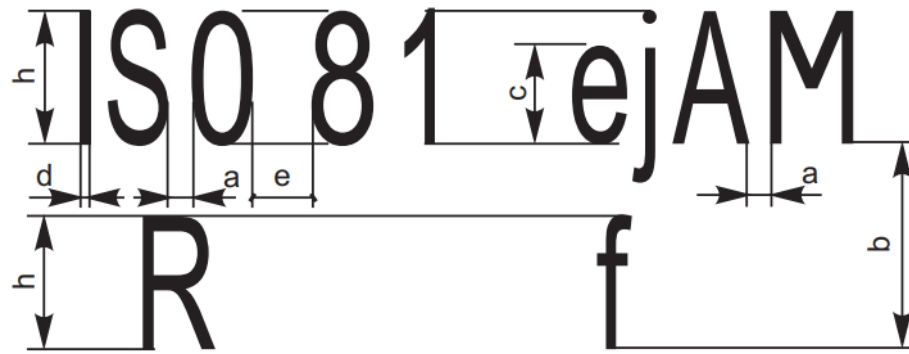
The following specifications are given for the dimensions of letters and numerals:

- I. The height of capital letters is taken as the base of dimensioning.
- II. The two standard ratios for d/h , **1/14** and **1/10** are the most economical, as they result in a minimum number of line thicknesses.
- III. The lettering may be inclined at **15°** to the **right**, or may be vertical.



Lettering Style

Cont'd



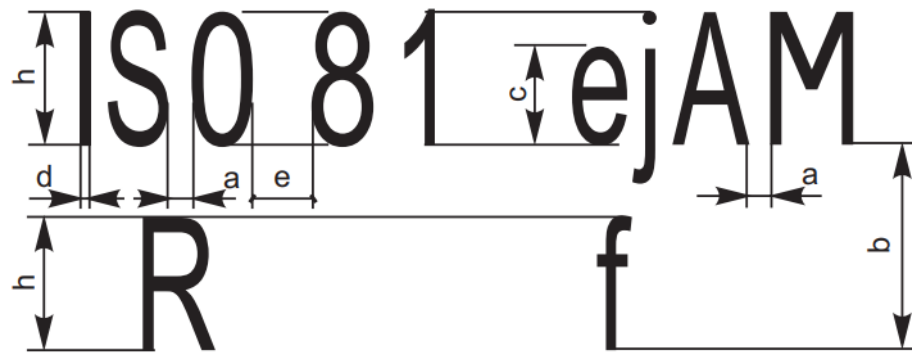
Lettering A ($d = h/14$)

Characteristic	Ratio	Dimensions, (mm)							
Lettering height (Height of capitals)	h (14/14) h	2.5	3.5	5	7	10	14	20	
Height of lower-case letters (without stem or tail)	c (10/14) h	—	2.5	3.5	5	7	10	14	
Spacing between characters	a (2/14) h	0.35	0.5	0.7	1	1.4	2	2.8	
Minimum spacing of base lines	b (20/14) h	3.5	5	7	10	14	20	28	
Minimum spacing between words	e (6/14) h	1.05	1.5	2.1	3	4.2	6	8.4	
Thickness of lines	d (1/14) h	0.18	0.25	0.35	0.5	0.7	1	1.4	

NOTE The spacing between two characters may be reduced by half, if this gives a better visual effect as for example LA, TV; it then equals the line thickness.

Lettering Style

Cont'd



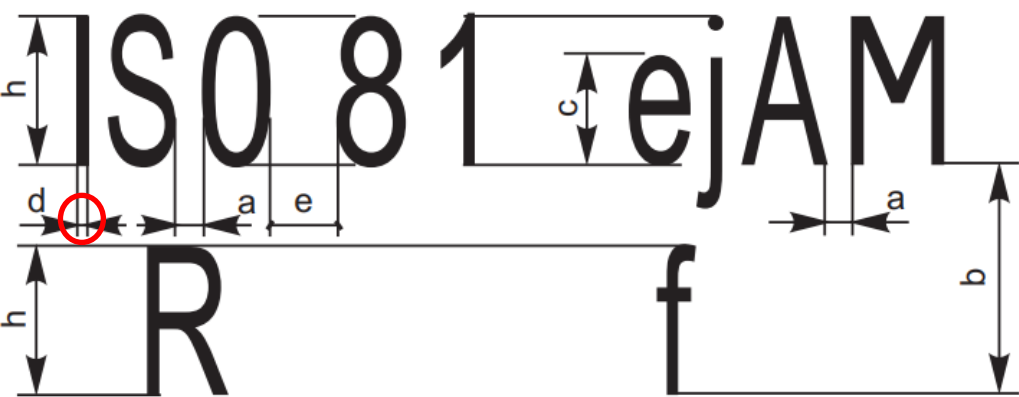
Lettering B ($d = h/10$)

<i>Characteristic</i>	<i>Ratio</i>	<i>Dimensions, (mm)</i>							
Lettering height (Height of capitals)	h (10/10) h	2.5	3.5	5	7	10	14	20	
Height of lower-case letters (without stem or tail)	c (7/10) h	—	2.5	3.5	5	7	10	14	
Spacing between characters	a (2/10) h	0.5	0.7	1	1.4	2	2.8	4	
Minimum spacing of base lines	b (14/10) h	3.5	5	7	10	14	20	28	
Minimum spacing between words	e (6/14) h	1.5	2.1	3	4.2	6	8.4	12	
Thickness of lines	d (1/10) h	0.25	0.35	0.5	0.7	1	1.4	2	

Letter Width Proportions

Uppercase Letters

Letter	Width (in multiples of d)
I	1d
J, L	5d
C, E, F, G, S, T, U, V, Y, Z	6d
B, D, H, K, N, P, R, X	7d
A, M, O, Q	8d
W	10d

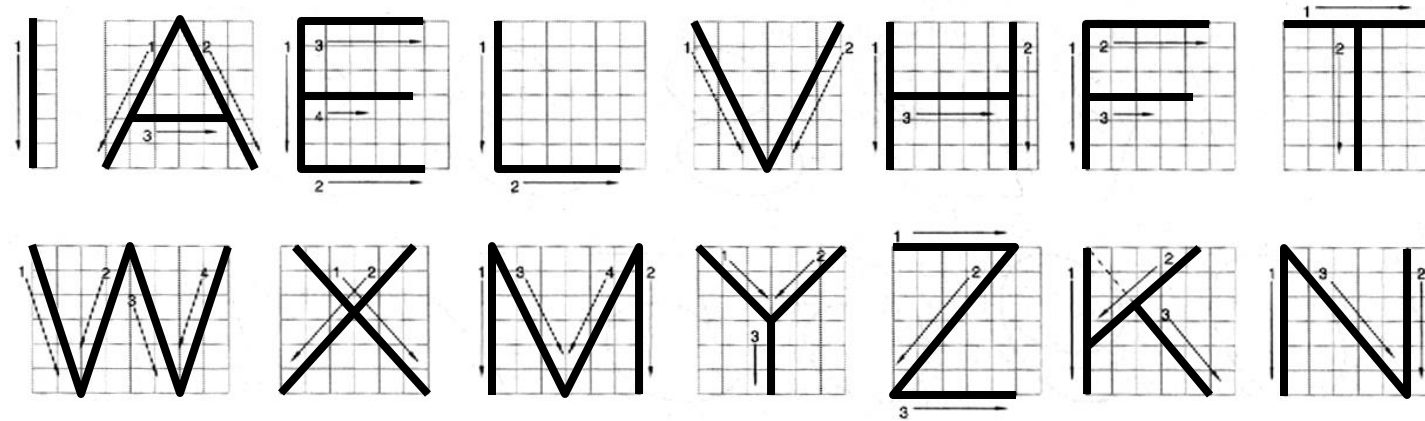


Numbers

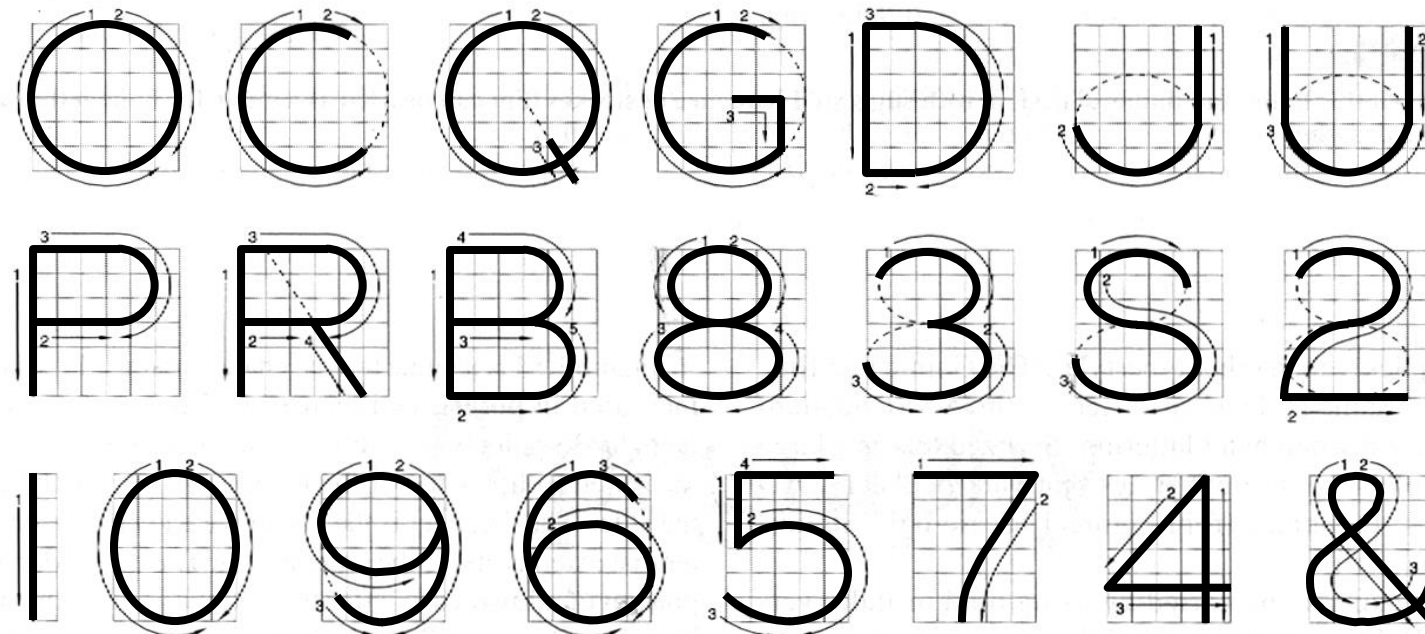
Number	Width (in multiples of d)
1	1d
2, 3, 5, 7	6d
0, 4, 6, 8, 9	7d

Upper-case Letters & Numerals

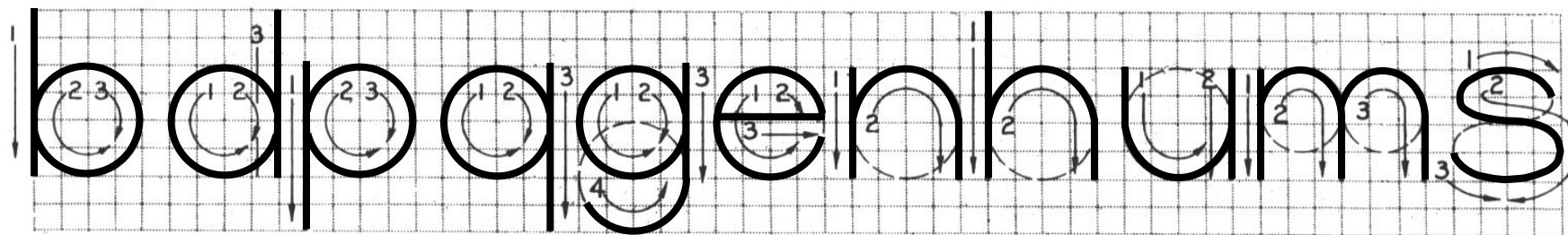
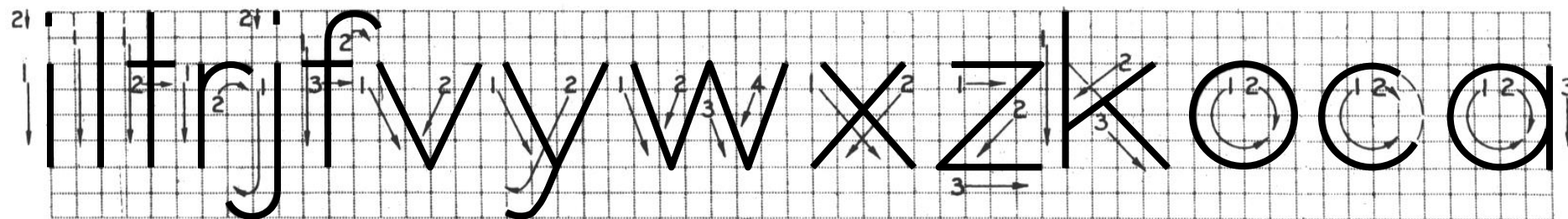
Straight line
letters



Curved line
letters
&
Numerals

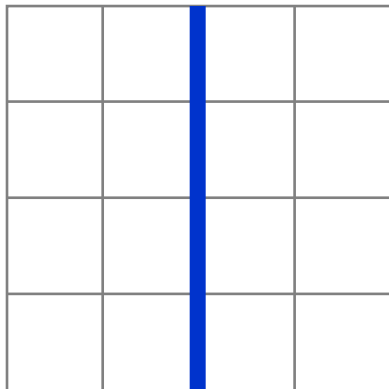


Lower-case Letters

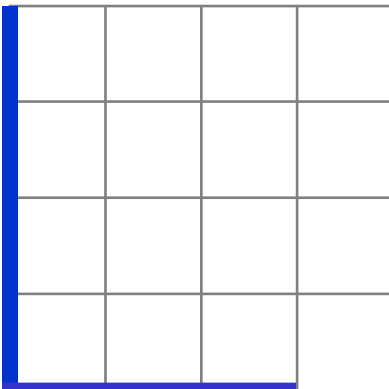


Stroke Sequence : Upper-case

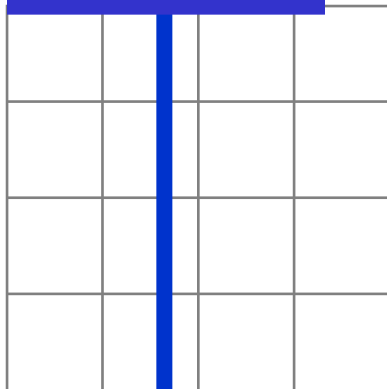
I



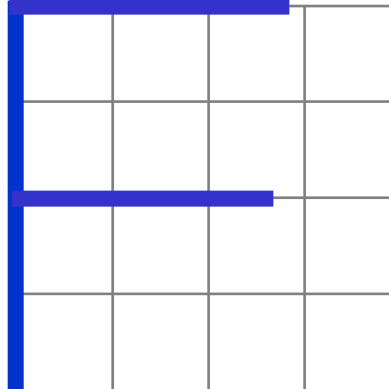
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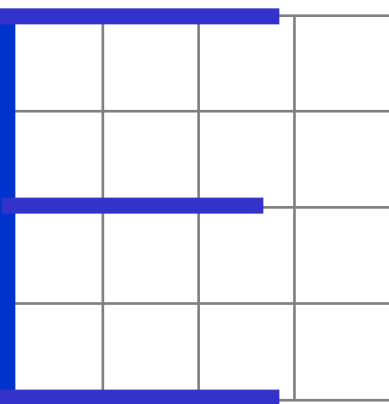
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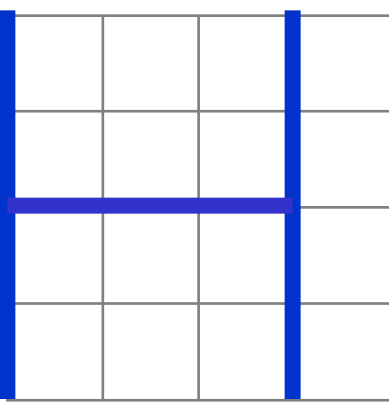
F



E

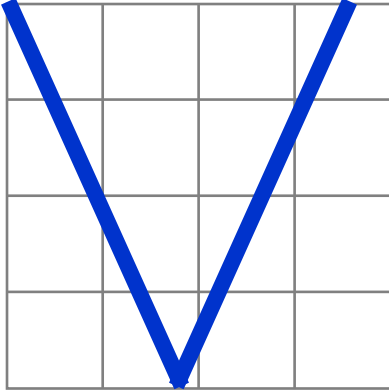


H

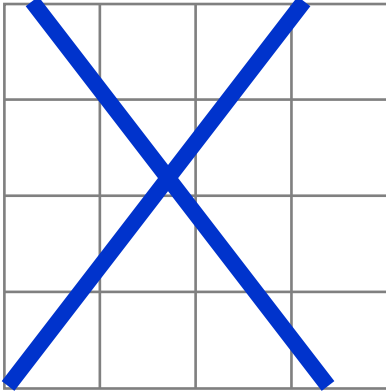


Stroke Sequence : Upper-case

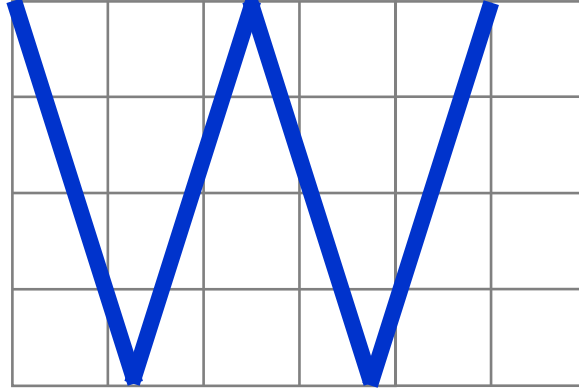
V



X

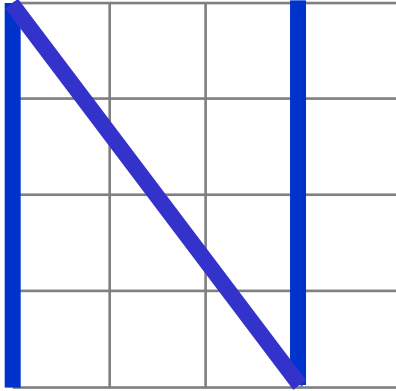


W

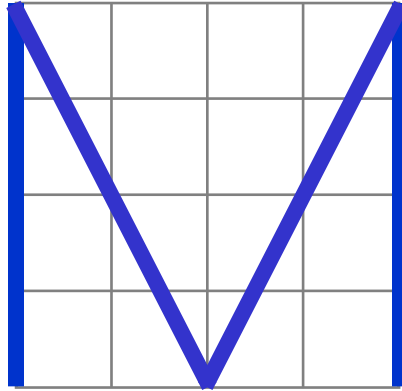


Stroke Sequence : Upper-case

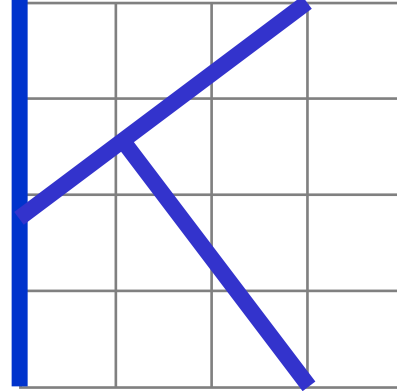
N



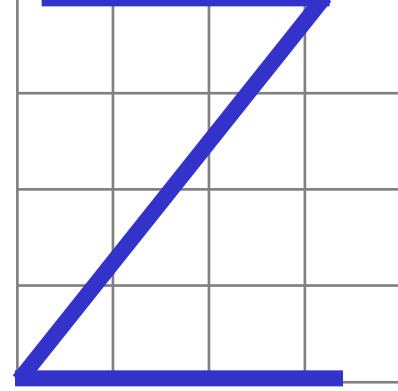
M



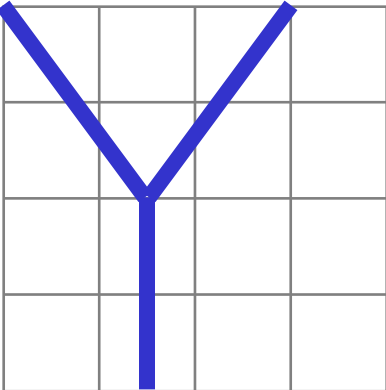
K



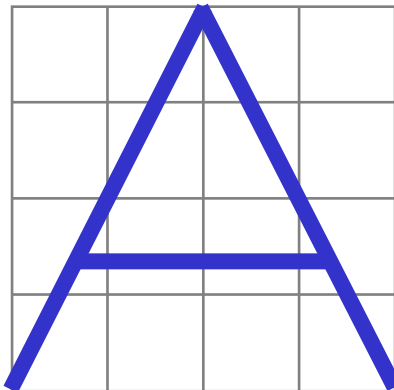
Z



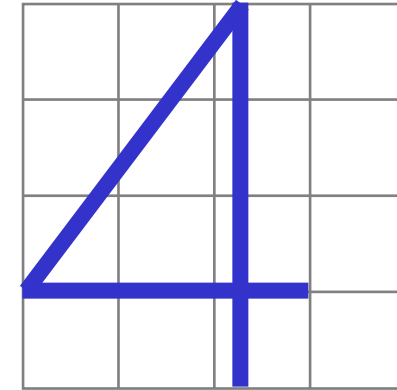
Y



A

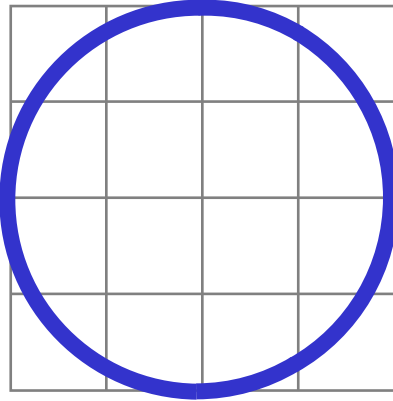


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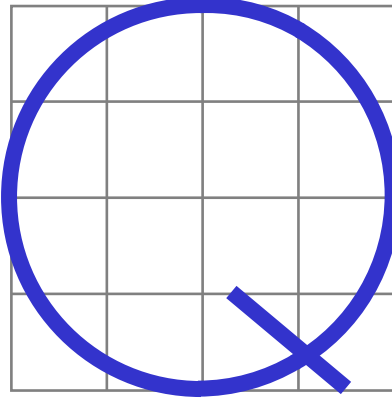


Stroke Sequence : Upper-case

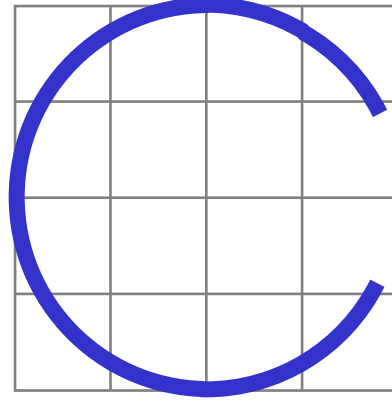
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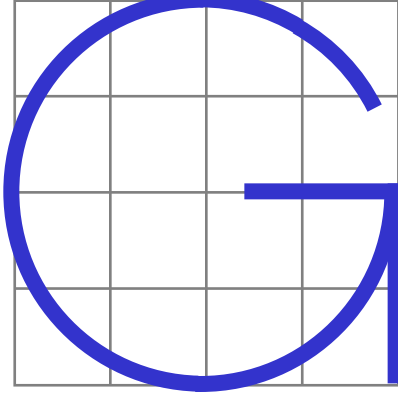
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C

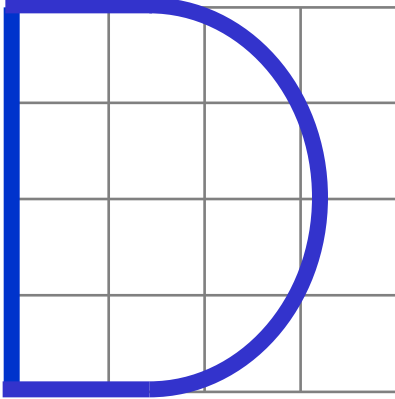


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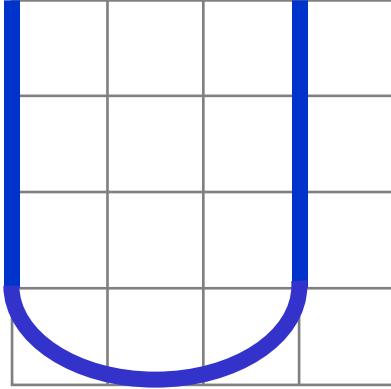


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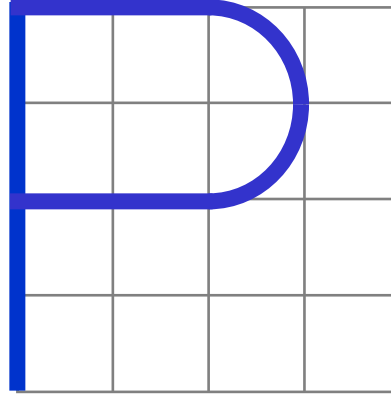
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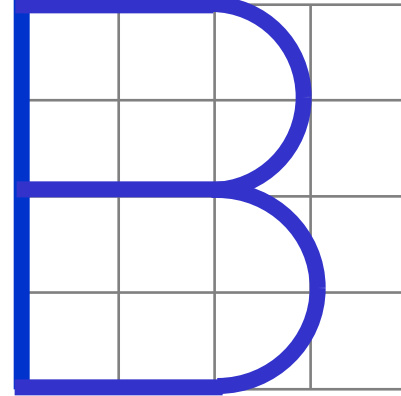
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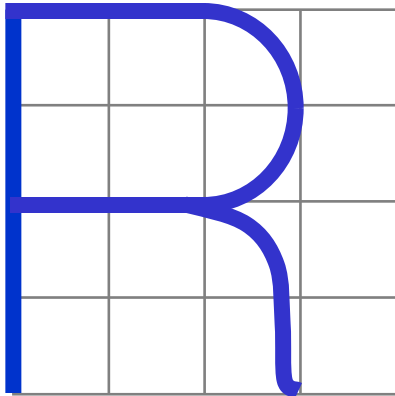
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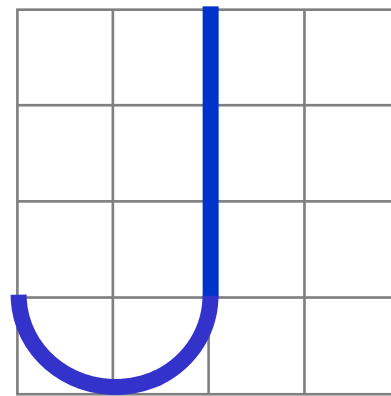
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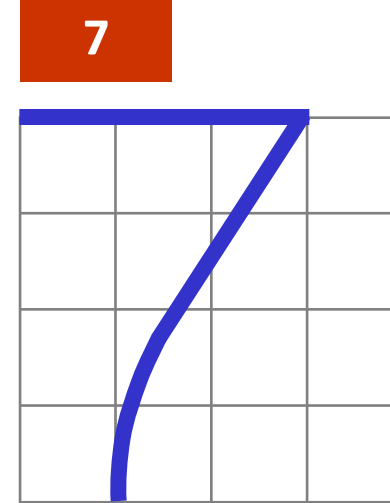
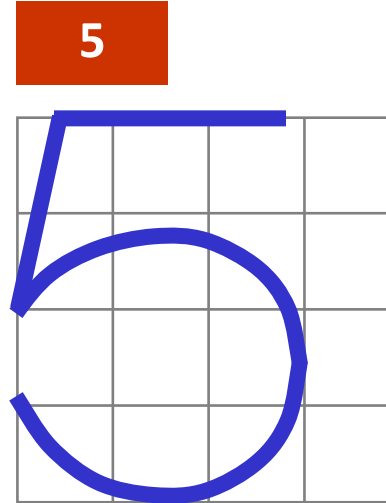
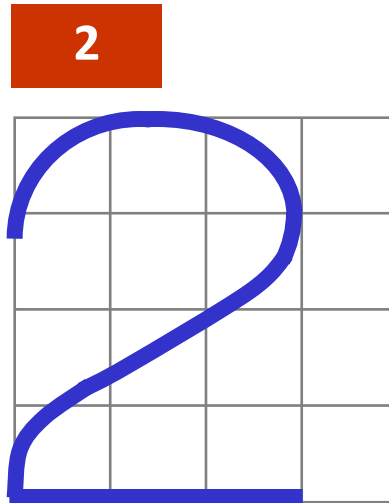
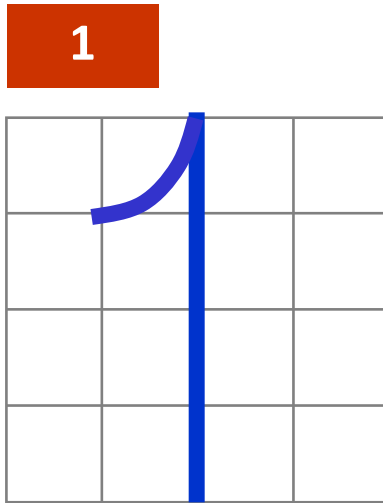
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J

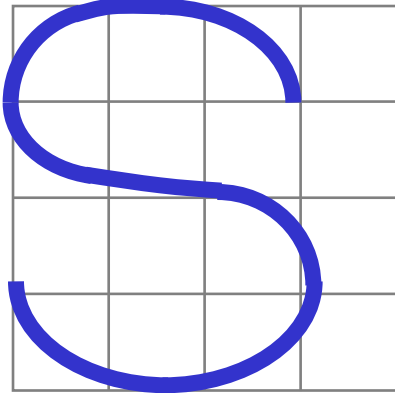


Stroke Sequence : Upper-case

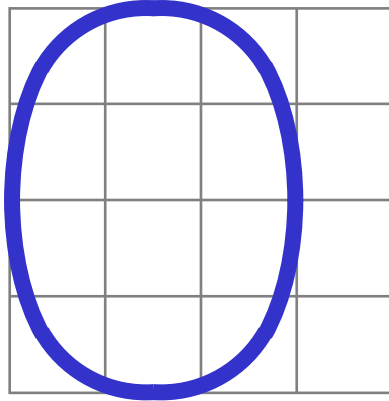


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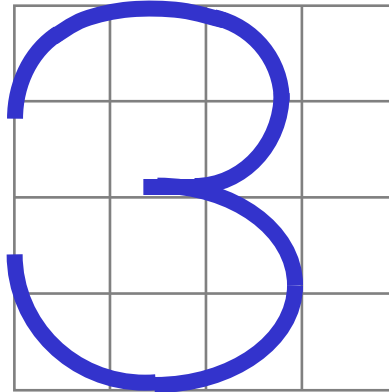
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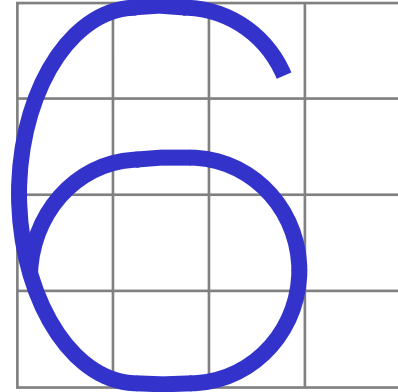
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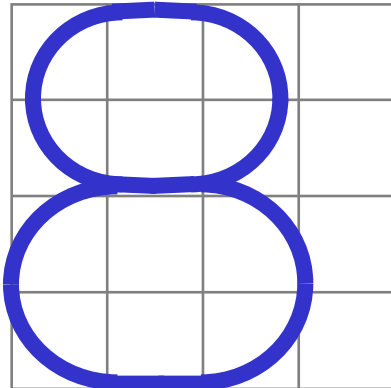
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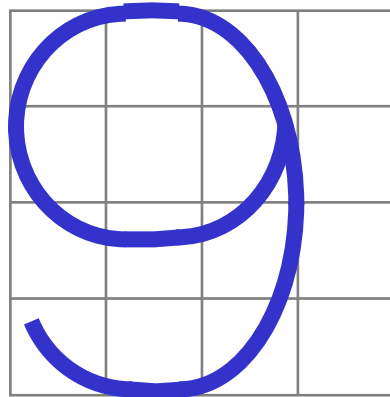
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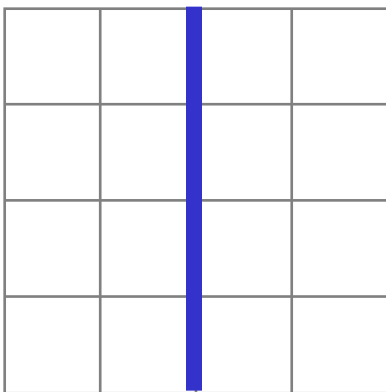


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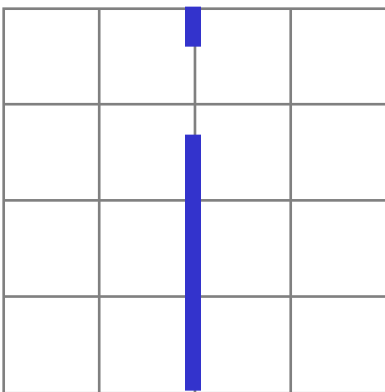


Stroke Sequence : Lower-case

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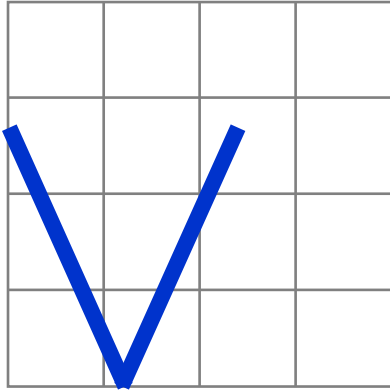


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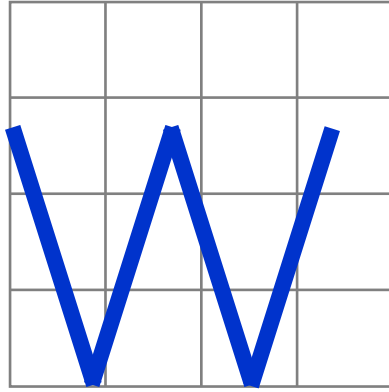


Stroke Sequence : Lower-case

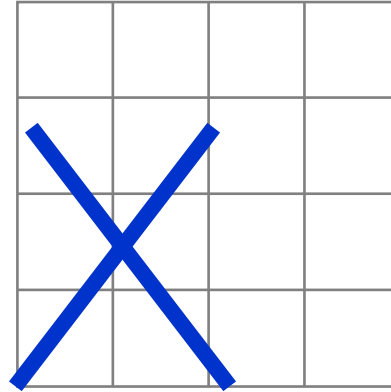
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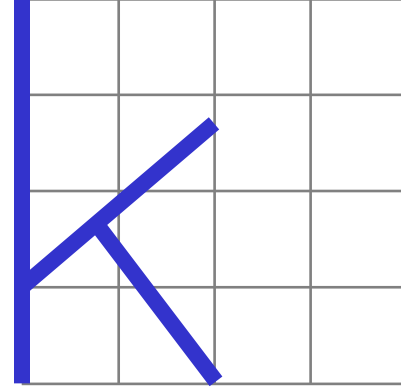
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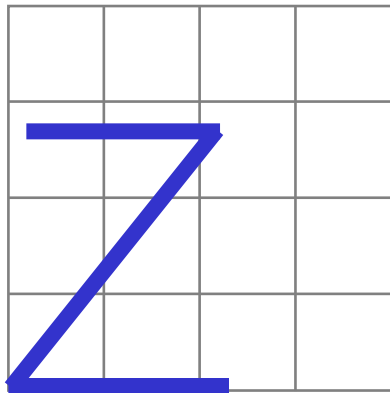
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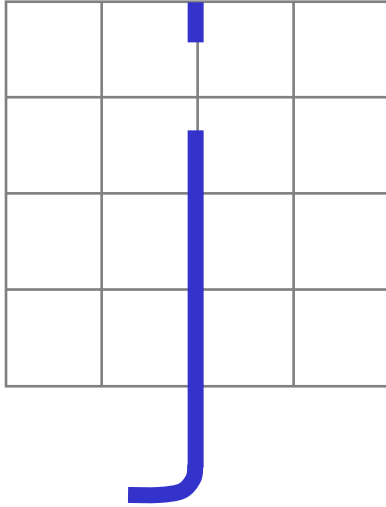


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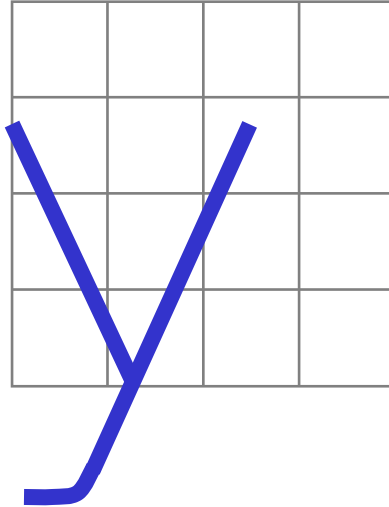


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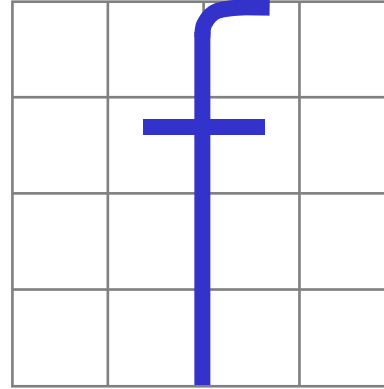
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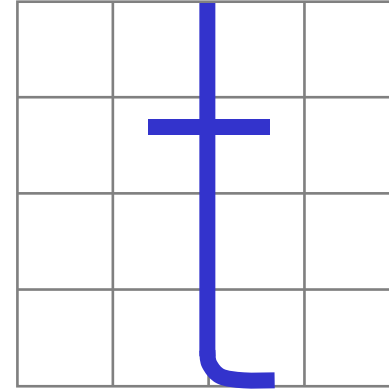
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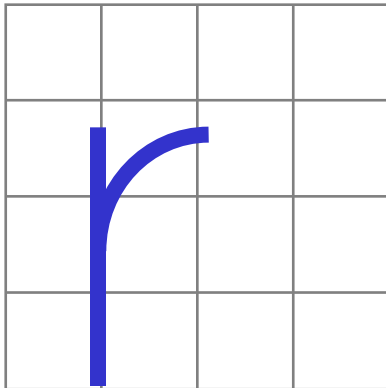
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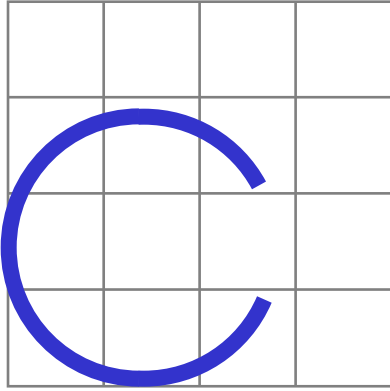


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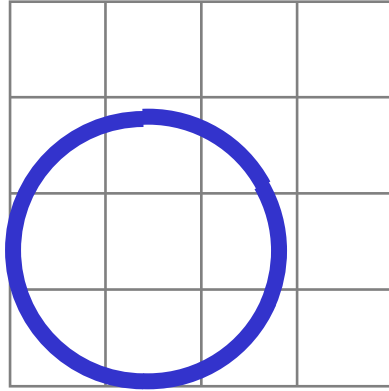


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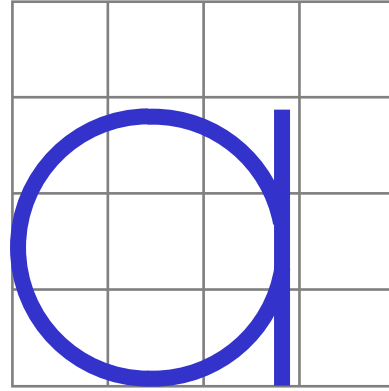
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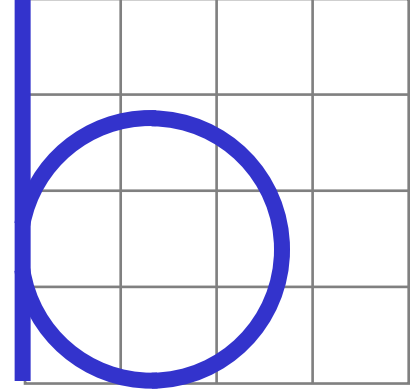
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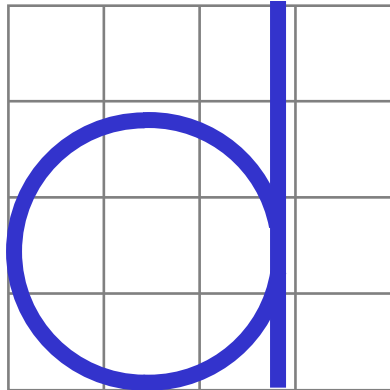
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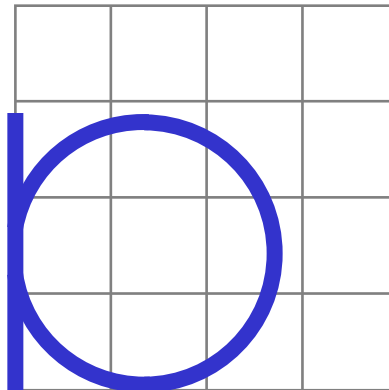
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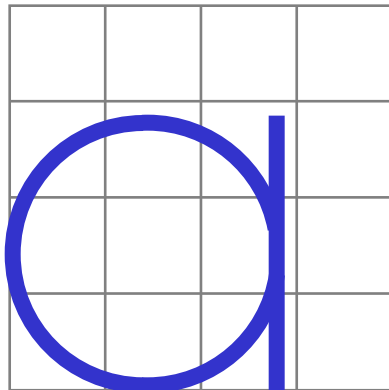
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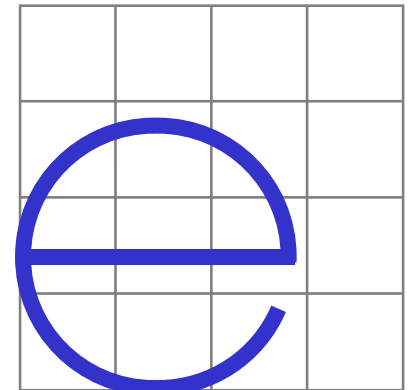
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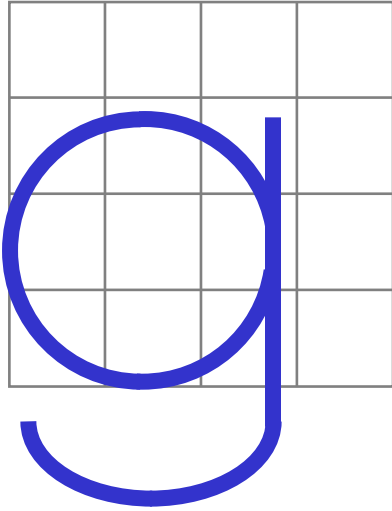


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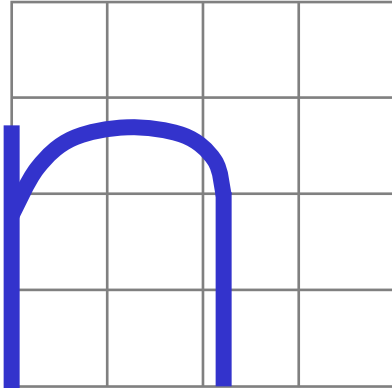


Stroke Sequence : Lower-case

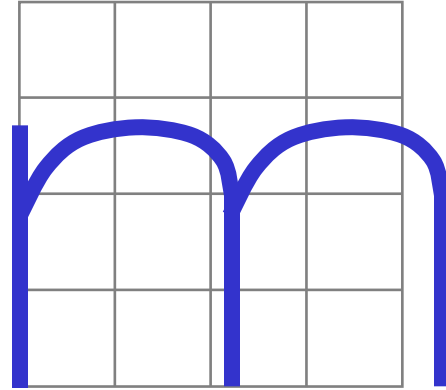
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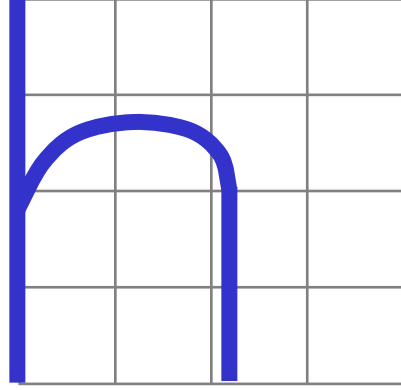
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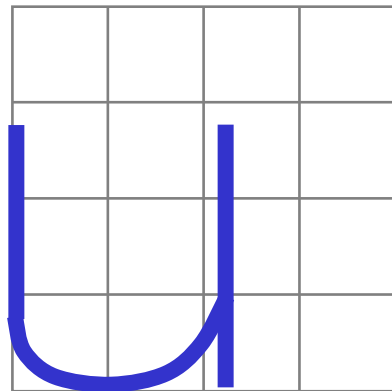
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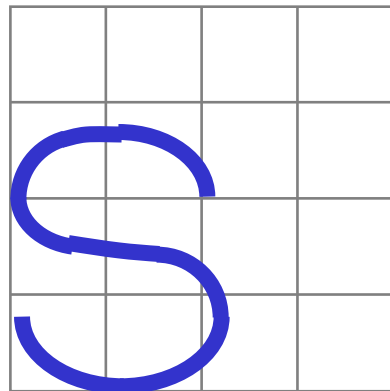
h



u



s



Word Composition

- Space between the letters depends on the adjacent contour of the letters.

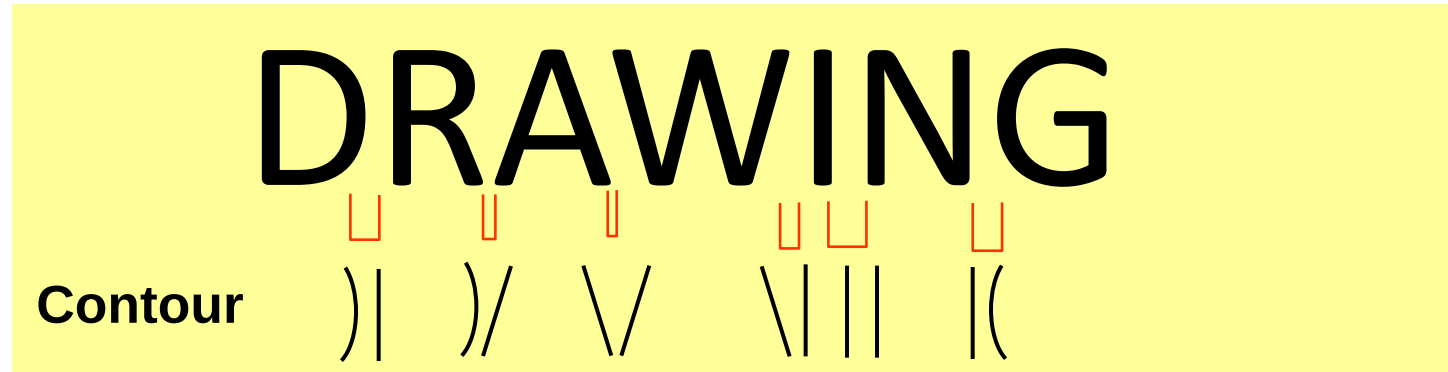
Non-uniform
spacing

DR A WI NG

Uniform
spacing

D R A W I N G

Word Composition



■ Contour can be denoted as “straight”, “slant” and “curve”.

■ Adjacent contour can be

1. straight-straight : II, IN, IM, IP etc.
2. straight-curve (or curve-straight) : IO, QR etc.
3. straight-slant (or slant-straight) : IV, IW etc.
4. curve-curve : OO, OG etc.
5. slant-curve (or curve-slant) : VO, WG, VC etc.
6. slant-slant : VW, VX etc.

Spacing

Sentence Composition

- Leave the space between words equal to the space requires for writing a letter “O”.

Example

ALL O DIMENSIONS O ARE O IN
MILLIMETERS O UNLESS
OTHERWISE O SPECIFIED.

Title Block

A **title block** is a standardized area on a technical drawing that provides essential information about the drawing itself. It typically appears in the bottom right corner or at the top of the drawing. Key elements included in a title block are:

- **Drawing Title:** A descriptive name of the drawing or part.
- **Drawing Number:** A unique identifier for the drawing, often part of a larger project or series.
- **Revision Number:** Indicates the version of the drawing, reflecting any changes made since the original.
- **Date:** The date the drawing was created or revised.
- **Author:** Name of the person or organization that created the drawing.
- **Scale:** The ratio of the drawing size to the actual size of the object.
- **Company Logo:** Branding of the organization responsible for the drawing.
- The title block serves as a quick reference for anyone reviewing the drawing, ensuring they have the necessary context and information about the specific part or assembly being represented.

Title Block

Cont'd

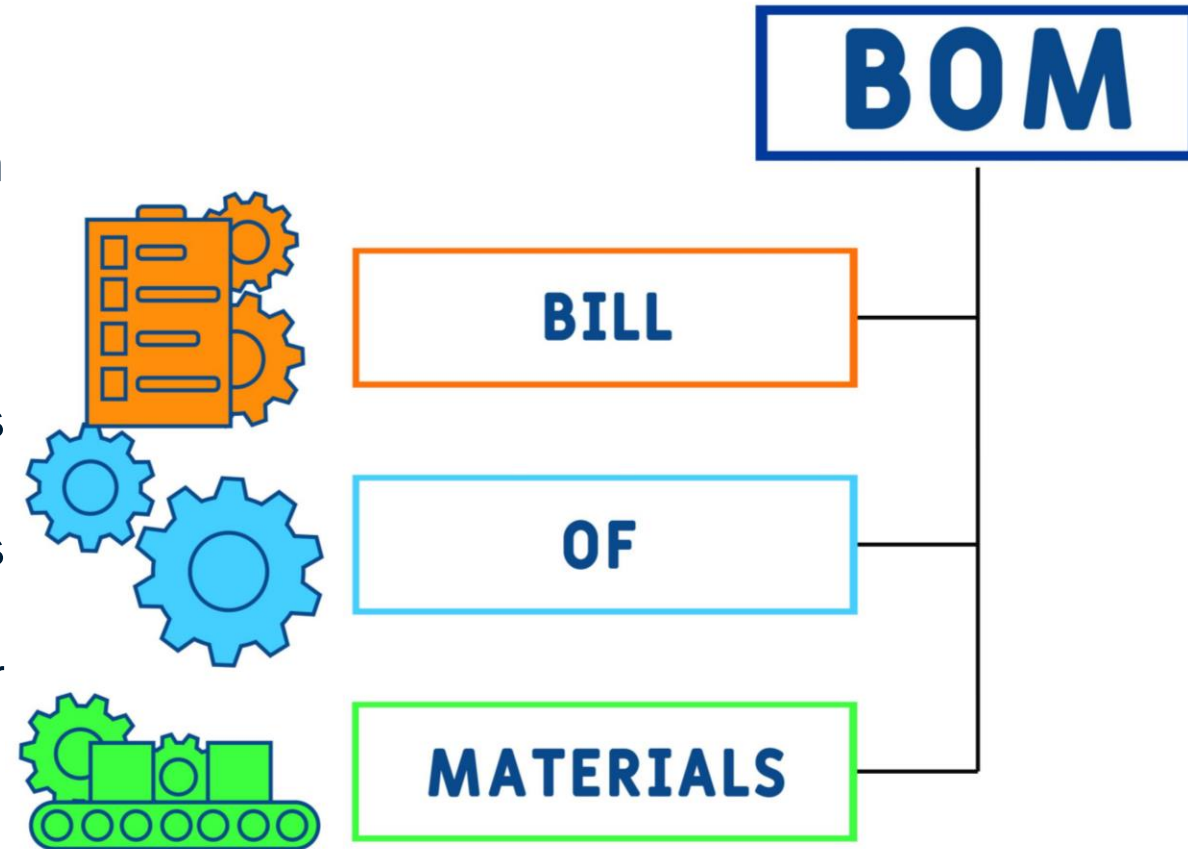
185					
50		70		35	
DRN. BY: ABCDEFG HIJKLMNOPQ RSTUVWXYZ				SCALE: 1:1, 1:2..	
ID. NO. 01234567 89 AND SEC. A, B, C, ...1, 2, 3...				DR.No. 1, 2, 3..	
CHKD BY: ADDISU A.				TITLE: MULTIVIEW or PICTORIAL or AUXILIARY VIEW or SECTIONAL VIEW or name of any part or assembly drawing of tool or machine.....	
FAP or TAP symbol		FULL NAME OF THE SCHOOL or INSTITUTE e.g BAHIR DAR UNIVERSITY BAHIRDAR INSTITUTE OF TECHNOLOGY FACULTY OF MECHANICAL AND INDUSTRIAL ENGINEERING		DIMENSION: mm	
				DATE: 02/ 10/ 2025	

Bill of material

- A **Bill of Materials (BOM)** is a **comprehensive, hierarchical** list of all **raw materials, components, sub-assemblies**, and **quantities** needed to **manufacture** an **end-product**, serving as a critical **blueprint** for **planning, costing, and production**.
- It acts as a central information source, guiding procurement and production to ensure efficiency, consistency, and accuracy in the manufacturing process.

A BOM typically contains:

- **Part Number/Code:** A unique identifier for each component.
- **Description:** A text description of the part or material.
- **Quantity:** The number of each component required.
- **Lifecycle Status:** Information on the component's development stage.
- **Approval Status:** Confirmation that the component has been approved.
- **Supplier Information:** Details about the vendors or suppliers of the components.



Why BOMs Are Important Production Planning:

- They provide a clear guide for assembly and the supply chain, ensuring all necessary materials are available at the right time.

Cost Control:

- By detailing all components and their quantities, BOMs are essential for product costing and minimizing expenses.

Quality Assurance:

- They ensure product consistency by dictating the exact components needed for manufacturing.

Efficiency and Reduced Waste:

- A well-structured BOM helps prevent production delays, cost overruns, and material waste.

Communication:

- BOMs serve as a universal reference point, facilitating clear communication between design, manufacturing, and supply chain teams.

Bill of material

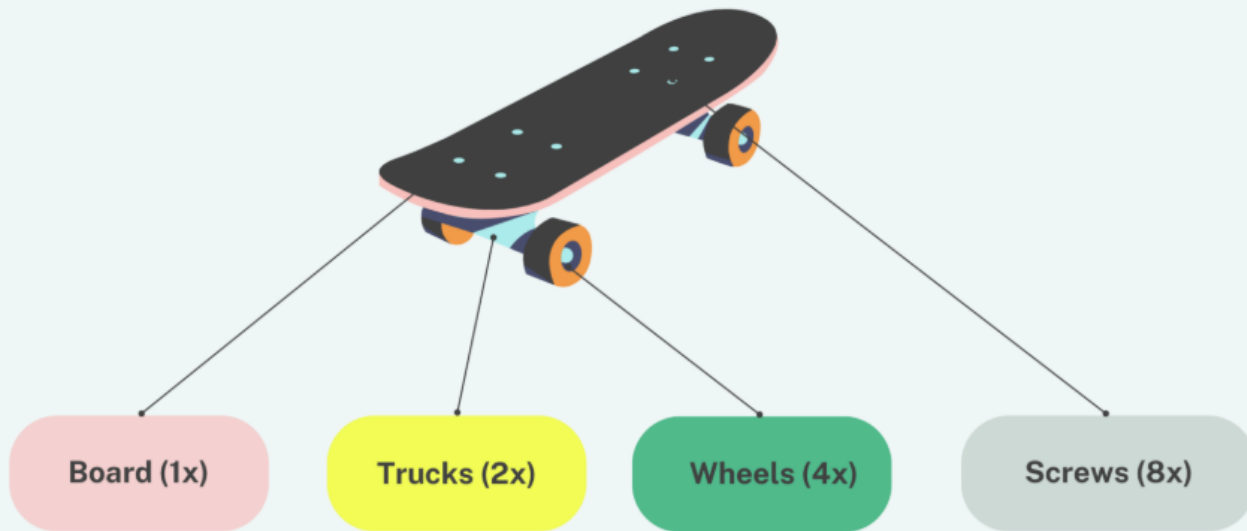
Cont'd

Types of BOMs

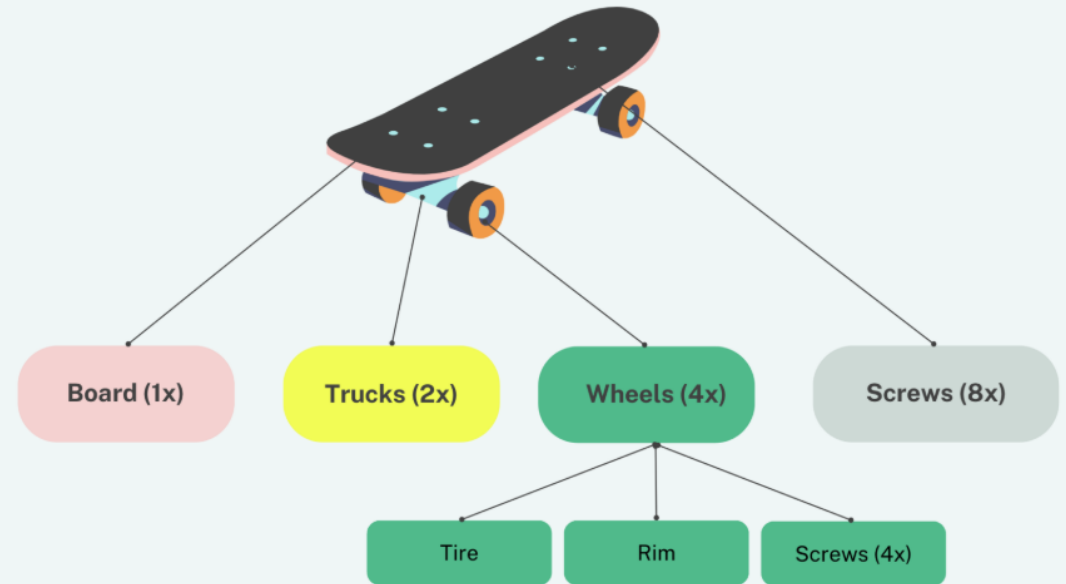
- While there are several specific types, BOMs can broadly be categorized by their hierarchical structure:

1. Single-Level BOM: Lists only the direct components and materials for a single assembly.

2. Multi-Level BOM: Shows the entire product structure, including sub-assemblies and their respective components in a hierarchical format.



Single-Level BOM

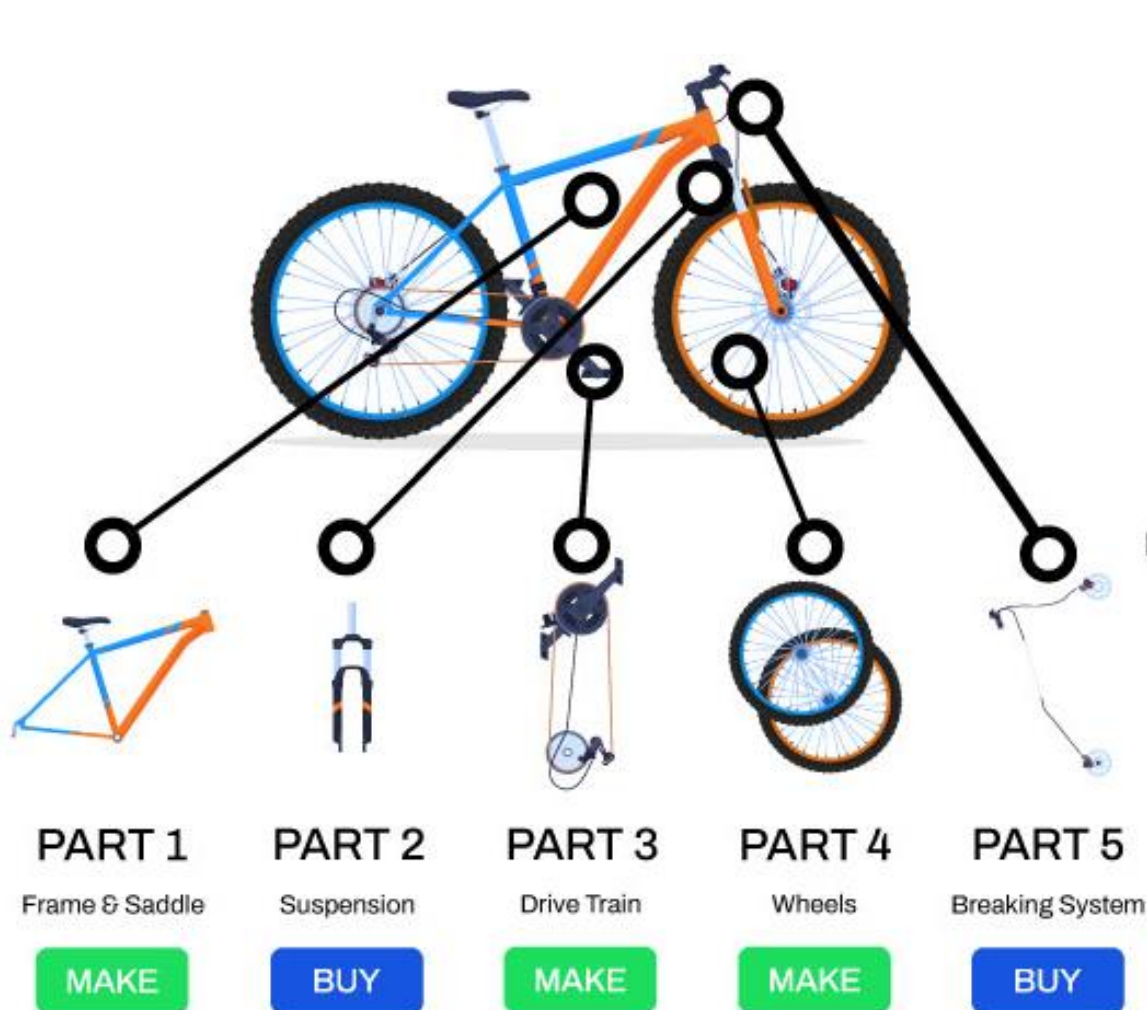


Multi-Level BOM

Bill of material

Cont'd

Parts of a Mountain Bike



Bill of material

Cont'd



mBOM
(Manufacturing BOM)



EBoM
(Engineering BOM)



SBoM
(Sales Bills of Materials)



Production BoM



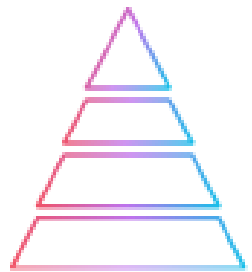
Service Bill of Materials



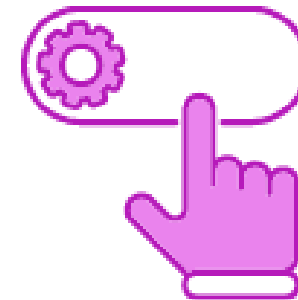
Assembly BoM



Single-level BoM



Multi-level BoM



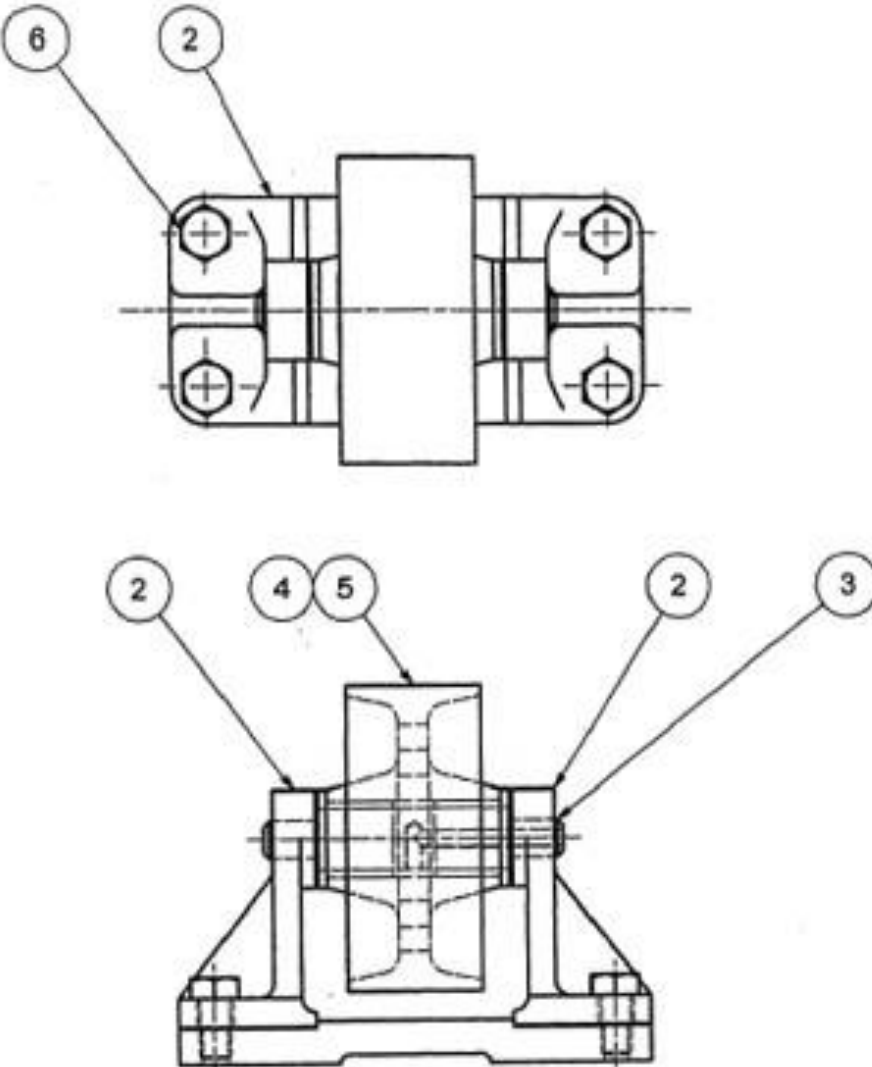
Configurable BoM



Template BoM

Bill of material

Cont'd



Technical drawing of a roller assembly. The drawing includes a front view, a side view, and an isometric view. The front view shows a central roller (5) mounted on a spindle (3) which is supported by two brackets (2). The brackets are mounted on a base (1) via bushings (4) and hex bolts (6). The side view shows the roller (5) and the spindle (3) in profile. The isometric view shows the entire assembly from a three-dimensional perspective.

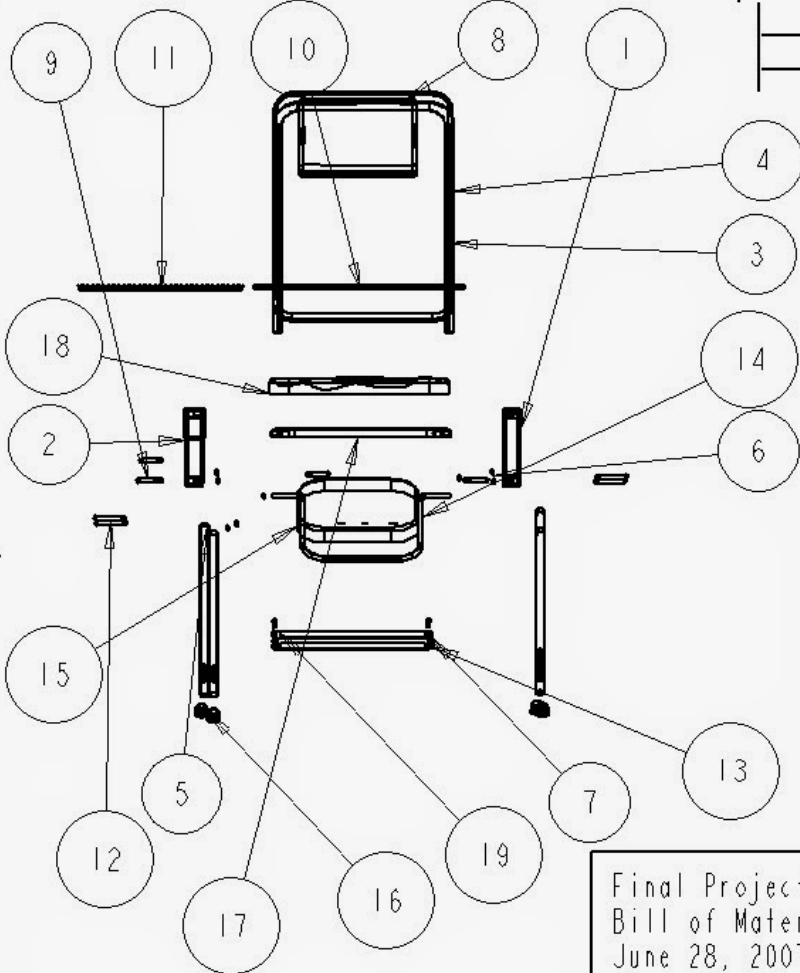
ITEM	DESCRIPTION	MATERIAL	QTY	PART NUMBER
6	Hex Bolt	Steel, Mild	4	ANSI B18.2.3.5M - M10 x 1.5 x 20
5	ROLLER	Cast Steel	1	RA_01-3
4	BUSH	Bronze, Soft Tin	2	RA_01-5
3	SPINDLE	Alloy Steel	1	RA_01-4
2	BRACKET	Cast Iron	2	RA01-2
1	BASE	Cast Iron	1	RA_01-1

MOUNT DRUITT COLLEGE OF TAFE
DETAIL DRAFTING

DRAWN	P.S.	TITLE
CHECKED		ROLLER ASSEMBLY
DATE	25:09:2009	SCALE 1:2
		DRG. NO. RA_01

Bill of material

Cont'd

			
Index	Part Name	Qty.	
1	ARMRESTL	1	
2	ARMRESTR	1	
3	BACK	1	
4	BACKPADDING	1	
5	LEG	4	
6	NUTARMRESTS	8	
7	NUTSUPPORT	4	
8	PILLOW	1	
9	PINARMRESTS	5	
10	PINBACK	1	
11	PINBACKSPRING	1	
12	PINLEGS	4	
13	PINSUPPORT	4	
14	RECLINER	1	
15	RECLINERPADDING	1	
16	RUBBERSTOPPER	4	
17	SEAT	1	
18	SEATPADDING	1	
19	SUPPORTBEAMLEGS	2	
Final Project Bill of Materials June 28, 2007		MAE377	
		32043713	

End of Chapter Two